

Viewpoint Invariance Is a Theorem, Not a Fit: An $SE(3)$ -Equivariant Recurrence for Rehabilitation Exercise Assessment

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Abstract—Automated rehabilitation scoring from skeleton sequences is evaluated almost exclusively on clean, frontal recordings — a regime in which architectures are indistinguishable and deployment failures are invisible. We show both halves of that claim on KIMORE. First, clean accuracy decides nothing: a 4.91M-parameter point-cloud transformer (PCT), a 0.66M-parameter $SE(3)$ -equivariant recurrence, and a 0.21M-parameter GRU on hand-crafted $SO(3)$ invariants score 6.47 ± 0.20 , 6.73 ± 0.30 and 6.31 ± 0.16 MAD — every gap inside the 0.33 MAD spread that training nondeterminism alone produces. A protocol audit shows the standard single-exercise slice of this benchmark carries *no subject-level signal at all* once epoch selection is made honest, so we pool and gate every experiment on a mean-predictor floor. Second, under deployment stress the models separate completely. Our read-out is $SE(3)$ -invariant *by construction* — a property of the architecture, certified numerically to 2.9×10^{-16} with a flat log-log slope in rotation angle (+0.160) and an 8.0×10^8 fp32/fp64 ratio, and holding for arbitrary weights. Rotating the test camera therefore moves our score by 9×10^{-6} MAD, while PCT decays past the mean predictor between 30° and 45° . Azimuth augmentation does rescue the baseline’s *aggregate* accuracy, but at a flat +1.0 MAD cost and with a per-sequence score shift of 3.03 MAD: a flat accuracy curve is a statistical fact about a population, not a guarantee about a patient. We report two negative results in full. Native consumption of irregular arrival times buys nothing here, and we derive why: a spectral census shows resampling destroys only 2.2% of KIMORE’s positional energy and acts as a denoiser, though the mechanism is real (grid-free tremor recovery holds at $r = 1.000$ under 70% drops where a resampled estimator decays to 0.785). And restoring chirality — closing a parity hole that made the invariant read-out silently $O(3)$ -invariant and provably blind to trajectory handedness — costs the viewpoint theorem nothing but buys no accuracy on a corpus of healthy-form exercises. What justifies the steerable encoder is measured, not argued: when a tracking node freezes, the hand-crafted invariant model collapses through the floor at a *single* dead joint ($6.31 \rightarrow 9.78$ MAD) while the graph-structured encoder degrades gracefully. Among models with an exact viewpoint guarantee, only the equivariant one survives a broken sensor — at $7.4\times$ fewer parameters than the baseline. Because the guarantee is architectural rather than learned, it is also co-designed for the edge: the read-out streams causally at 8.9 ms per frame (112 fps, a score before the exercise ends), and under int8 quantisation weight-rounding preserves the viewpoint theorem *exactly* while only activation-rounding perturbs the reported score — by 0.05 MAD, two orders below the clinical floor.

Index Terms—Equivariance, $SE(3)$, rehabilitation exercise assessment, viewpoint invariance, skeleton sequences, irregular sampling, real-time streaming inference, edge quantisation, time-to-first-score, KIMORE, NTU RGB+D, negative results.

I. INTRODUCTION

Automated assessment of physical rehabilitation exercises from skeleton sequences promises objective, remote, and continuous monitoring of therapy [1]. The dominant evaluation regime, however, is a narrow one. Models are trained and tested on clean, frontal recordings from a single sensor, and are ranked by a single aggregate error against a physician’s score [2]–[5]. That regime rewards capacity and penalises nothing that a deployed system would actually encounter: a camera placed against a different wall, a tracking node that stops updating, a dropped frame.

This paper is an argument that the regime, not the architectures, is the problem — and it begins by conceding the point the regime is supposed to settle. On clean, frontal KIMORE our proposed model **does not beat** the point-cloud transformer baseline. Neither does a GRU fed six lines of hand-crafted invariant features. The three are statistically indistinguishable, and we can say so precisely because we first measured what a difference has to exceed to be a difference: training on this corpus is not bit-reproducible (nondeterministic cuDNN and scatter-add kernels), and repeated fits of an *identical* configuration differ by ≈ 0.33 MAD — larger than every clean-accuracy gap between the leading models in this paper. Any single-run comparison on this benchmark is therefore reporting its own seed.

The situation is worse than a tie. Auditing the reference protocol on the single-exercise slice on which KIMORE results are most often reported, we find that the published practice of selecting the reported epoch by its test-set minimum is worth +1.111 MAD (16.8%) — and that once it is removed, the honest number is *statistically indistinguishable from a model that ignores its input entirely* (paired subject-cluster bootstrap $\Delta = -0.081$, 95% CI $[-0.803, +0.625]$). It is not subject leakage: on that slice each subject contributes one recording, so a random split is already subject-disjoint. The slice simply has no subject-level signal to measure. Pooling the five exercises restores it, and every experiment below is consequently pooled, subject-disjoint, and gated on provably clearing a per-exercise mean-predictor floor — because a model that ignores its input has *zero* degradation under any corruption and would win every robustness plot ever drawn.

With clean accuracy neutralised as a discriminator, the question becomes what a structural guarantee buys. We formulate assessment as an *invariant read-out of an equivariant representation*: a steerable message-passing encoder on the skeleton

graph produces per-joint features carrying $O(3)$ irreducible representations, an explicit projection onto a generating set of invariants discharges the symmetry, and a recurrence consumes the resulting invariant code together with the sensor’s actual inter-arrival times. Invariance of the predicted score under any rigid motion of the camera is then a theorem about the architecture (Prop. 2), true for arbitrary weights — trained, untrained, or adversarially chosen — rather than a behaviour induced by augmenting the training set.

The distinction is not scholastic, and the experiments are designed to make it bite. A rotation-augmented baseline has a *flat aggregate accuracy curve* across camera azimuth; by the metric the field reports, it has solved viewpoint. Its per-sequence degradation is 3.03 MAD: individual patients’ scores swing by three points of fifty as the camera moves, and the errors merely cancel in the mean. Ours move by 9×10^{-6} . A screening instrument is required to return the same patient the same score from a different room, and that is a statement about each prediction, not about a population average. We therefore report degradation and accuracy side by side throughout, and neither alone.

We also report, in full, the two places where the structural story does *not* pay. Consuming irregular arrivals natively — the standard motivation for continuous-time models — confers no advantage on KIMORE; the fixed-grid baseline is better on both accuracy and stability at every drop level, under an identical hash-locked augmentation budget. Rather than excuse this, we derive it: a model-free spectral census shows 97.7% of the corpus’s positional energy lies below the baseline’s resampling corner, so the resampling operator destroys 2.2% of the signal and mostly discards differencing-amplified sensor noise. It is a *denoiser*, and the null was predictable before the experiment was run. That the mechanism nevertheless exists we confirm by placing signal in the destroyed band: a physiologically calibrated 5 Hz tremor is recovered from the true irregular stamps at $r = 1.000$ under drop rates up to 70%, while a resampled estimator decays to 0.785. The claim is a scope boundary, not a defeat. Likewise, closing the model’s parity hole — the invariant cut we and, to our knowledge, every comparable construction emit is composed entirely of parity-*even* features, and is therefore silently invariant to a mirror of the entire performance, hence blind to trajectory handedness — costs the viewpoint theorem nothing but earns no accuracy on a corpus of healthy-form prescribed exercises, which contains no chiral pathology to detect. We adopt the fix anyway, on the grounds that the architecture should not *assume* handedness is irrelevant, and we say plainly that it is unrewarded here.

Finally, the tie with the hand-crafted invariant model leaves an obligation that elegance cannot discharge: if a plain GRU on named geometric features is exactly as accurate *and* exactly as viewpoint-invariant, the steerable machinery is unjustified. We discharge it with a measurement. When Kinect tracking nodes freeze — the characteristic failure of a consumer depth sensor — the hand-crafted model does not degrade gracefully; it fails outright: a *single* dead joint drives it from 6.31 to 9.78 MAD, through the mean-predictor floor, because its features *name* the joints they depend on. The graph encoder pools learned

messages over the skeleton and dilutes the same failure across 25 joints. This is invisible in every clean-accuracy column in the paper, which is exactly why it had to be measured separately.

Finally, we treat the viewpoint guarantee as a *systems* property, not only a mathematical one: we audit its precision budget under aggressive integer quantisation and convert the read-out to true frame-by-frame causal streaming, and find that geometric exactness survives to int8 — where only activation-rounding, not weight-rounding, perturbs it — and runs strictly causally at 8.9 ms per frame with no full-clip buffering, so a structural safety guarantee and low-power, real-time operation are co-designed rather than traded off. This is the third axis on which architecture, and not accuracy, is the decisive variable.

Our contributions are:

- 1) **A protocol audit that establishes what the benchmark can measure.** Optimistic epoch selection inflates the standard KIMORE single-exercise result by 16.8%, and the honest number does not beat a mean predictor. We identify the mechanism (it is *not* subject leakage), fix a pooled protocol, quantify the 0.33 MAD non-determinism floor below which no accuracy difference is interpretable, and gate every subsequent claim on clearing the floor.
- 2) **Exact viewpoint invariance, certified rather than asserted.** An $SE(3)$ -invariant read-out over a steerable encoder, audited by eight numerical gates (E1–E8): representation orthogonality $\leq 10^{-12}$, violation at machine roundoff with a *flat* log-log slope in rotation angle (+0.160), an fp32/fp64 violation ratio of 8.0×10^8 , and translation invariance at 2.6×10^{-16} . Magnitude alone would not license the word “zero”; flatness and precision-scaling exclude a small modelling violation masquerading as noise. Measured cross-viewpoint degradation is 9×10^{-6} MAD against the baseline’s 9.42, and the baseline crosses the mean-predictor floor between 30° and 45° . The guarantee is not confined to a synthetic sweep: on NTU RGB+D 60 Cross-View — a real cross-camera benchmark whose test recordings come from a 45° -displaced camera — the identical encoder, trained without rotation augmentation, generalises to the held-out camera at exact (5.8×10^{-9} , fp64) cross-view logit invariance.
- 3) **A co-designed deployment path: a structural quantisation audit and a causal streaming read-out.** The guarantee is not merely an fp64 identity, and we engineer it to hold on the hardware a home device actually runs. Auditing its precision budget from fp64 to int8, we find the theorem survives *weight* quantisation exactly (equivariance holds for arbitrary weights, so a quantised model is simply a different equivariant model), and that the $\times 10^5$ floor cliff is attributable *solely* to activation grid-rounding of the steerable channels — a structural, not a weight-precision, phenomenon — while the reported clinical score still moves only 0.05 MAD, below the floor. And because the invariance lives entirely in a per-frame encoder and an invariant cut, the read-out converts to a strictly causal, unidirectional form with a running-

mean pool that emits a first score 8.9 ms after the first frame (112 fps) and preserves invariance to 1.4×10^{-8} , giving the architecture an honest time-to-first-score for real-time use (Sec. V). The theorem is engineered to hold from fp64 to int8 and from full-clip to first-frame, not only in exact arithmetic on a full recording.

- 4) **The capacity tax of learned invariance, and the sense in which a flat curve is not invariance.** Azimuth augmentation does flatten the baseline’s aggregate response, at a flat +1.0 MAD penalty that parks it *at* the floor — and it leaves a 3.03 MAD per-sequence score shift that aggregate MAD cannot express. We grant the baseline this remedy explicitly rather than defeat a strawman.
- 5) **A parity hole in invariant skeleton read-outs, exhibited and closed.** Gate E6 shows the parity-even cut returns *bit-identical* output on a mirrored body: an unintended $O(3)$ -invariance. We restore $SO(3)$ with pseudo-scalar triple products and anatomical signed volumes, re-certify read-out invariance at 2.9×10^{-16} (the fix is free), and separately measure — against the natural objection — that reflection-blindness was never left/right-blindness (gate E7).
- 6) **Two negative results, one of them derived.** Grid-free consumption of arrival times buys nothing on KIMORE, and a spectral census predicts exactly that (2.2% of positional energy destroyed; the mechanism verified live at $r = 1.000$ tremor recovery). Chirality is restored but unrewarded. We state both as scope boundaries with the corpus properties that would reward them.
- 7) **A measured justification for the steerable encoder: sensor-node failure.** Among models carrying an exact viewpoint guarantee, only the graph-structured one survives a broken tracking node (+3.76 vs. +12.05 MAD over 8 dead joints). We also record, against ourselves, that the point-cloud transformer is the most node-robust model of the three — robustness to node failure is not a property of equivariance and we do not claim it is. The defensible claim is the Pareto one: our model is the only arm that clears the floor under *both* a 90° camera move and two dead joints while retaining a per-prediction guarantee, at $7.4\times$ fewer parameters than the baseline.

II. RELATED WORK

A. Rehabilitation Exercise Assessment from Skeletons

KIMORE [1] is the standard benchmark: subjects perform five trunk and hip exercises, each scored 0–50 by a clinician. Reported results span a wide range of architectures — ST-GCN variants with supervised contrastive objectives [3], cross-modal augmentation [4], dual-stream spatio-temporal graph networks [5], and, most relevantly here, point-cloud transformers that treat the skeleton as an unordered set of 3D points with axial self-attention [2]. Ismail-Fawaz et al. [6] aggregate KIMORE, UI-PRMD [7] and IRDS under cross-subject splits and report error rather than rank correlation, and their protocol discipline is the closest antecedent of our own audit.

Two properties of this literature motivate the present work. First, evaluation is uniformly in-distribution: same sensor,

same camera placement, no simulated hardware failure. Second, protocols differ enough that headline numbers are not comparable — a point we make quantitative in Sec. IV-A, where we isolate optimistic epoch selection as a 16.8% effect on the most commonly reported slice, and where we find that slice to carry no subject-level signal at all. To our knowledge no prior work on this task reports a mean-predictor floor, and without one a degradation curve cannot be interpreted.

B. Equivariant and Invariant Architectures

Group-equivariant networks impose symmetry as a constraint on the hypothesis class rather than a target of fitting [9]. For 3D geometry the steerable family — tensor-field networks [10], $SE(3)$ -Transformers [11], and the $e3nn$ framework [12] — carries features in irreducible representations of $O(3)$ and constrains every layer to intertwine with the Wigner- D action, guaranteeing equivariance for arbitrary weights. Lighter constructions obtain $E(n)$ -equivariance without spherical harmonics [13], or lift a network to $SO(3)$ -equivariance by replacing scalar neurons with vector-valued ones [14]. We build the spatial encoder on [12], but the paper’s claim is deliberately *not* about steerable machinery: we show a GRU on hand-crafted $SO(3)$ invariants attains the same accuracy and the same exact viewpoint guarantee at a third of the parameters, and we then isolate — by measurement, in Sec. IV-E — the one axis on which the two genuinely differ.

An orthogonal line makes a non-equivariant backbone invariant by canonicalisation or frame averaging [15], which is exact but pays a forward-pass or a learned-canonicaliser cost. The two routes differ in where exactness comes from. A canonicaliser aligns each input to an estimated reference frame; the invariance is then exact *only in so far as the frame estimate is*, and a learned or ill-conditioned frame (e.g. under a near-degenerate pose or a symmetric skeleton) reintroduces the very per-prediction variance we set out to eliminate, while frame averaging over a rotation group is exact but multiplies the forward cost. Our invariant cut estimates no frame at all: it is an explicit projection onto a generating set of invariants of the latent, exact for arbitrary weights by Proposition 2, and Sec. III-C makes the resulting group precise — including the observation, which we have not seen stated, that the natural such cut is complete for $O(3)$ rather than $SO(3)$ and is therefore blind to trajectory handedness until pseudo-scalars are admitted. A head-to-head comparison against a canonicalisation front-end — on both aggregate accuracy and the per-sequence degradation of Sec. IV-C, where a frame-estimation error would surface — is a natural extension we do not run here.

C. Learned Invariance and Its Price

The default remedy for viewpoint variation is training-time augmentation, and it works in the sense the field measures: an augmented model’s aggregate error becomes approximately flat in camera angle. We grant the baseline exactly this remedy and report what it costs — a flat accuracy penalty, and a per-sequence score shift that survives the augmentation because augmentation makes errors *cancel* rather than *vanish*. The

distinction between a symmetry that holds in expectation over a training distribution and one that holds identically for every input is, in a clinical instrument, the whole point; it is expressible only if degradation is reported per sequence, which is why we report it alongside accuracy everywhere.

D. Irregular Sampling and Continuous-Time Models

Depth-sensor streams are not uniformly sampled: our anchor KIMORE recording has $C_V = 0.25$ in its inter-arrival times with genuine multi-frame dropouts. The standard treatment is to resample onto a uniform grid; the standard alternative is a continuous-time model that consumes arrival times natively — Neural ODEs [16], latent ODEs and ODE-RNNs for irregularly sampled series [17], Neural CDEs driven by an interpolant of the observations [18], or recurrences with explicit decay between observations [19]. We implement the Neural CDE variant, certify it to the same standard as the proposed model, and report it as a *control*: it satisfies an exact equivariance certificate and *fails* the mean-predictor floor, which is the cleanest demonstration we can offer that a structural guarantee is worthless without measured signal, and the reason our evaluation gates on the floor at all.

Our analysis of the adaptive solver (Sec. III-F) identifies a failure mode we have not seen documented: a Dormand–Prince controller [21] accepts or rejects steps on a scalar reduction of the embedded error, and the standard per-component scaled RMS norm is *not* group-invariant even when the vector field is exactly equivariant and the representation orthogonal. The rotated and unrotated integrations then receive different step grids and silently de-align. We prove the condition under which this happens and construct a per-irrep-normalised norm that removes it, driving grid divergence to identically zero.

Finally, the negative result of Sec. IV-D is stated as a bandwidth condition and verified in both directions, the constructive half using the classical fact that irregularly sampled data have no single Nyquist frequency and do not alias — the basis of the Lomb–Scargle construction [22] — which is precisely why a grid-free estimator recovers a 5 Hz tremor that a uniform grid at the same average rate folds onto itself irrecoverably.

E. Robustness to Missing and Corrupted Joints

Occlusion and tracking failure are standard concerns in skeleton-based action recognition, typically addressed by masking-based pretraining or by attention that can down-weight anomalous joints [8]. We adopt the failure literally — a Kinect node that stops updating and reports a stale estimate — and use it as the discriminating experiment between two models that are otherwise indistinguishable in accuracy and in viewpoint guarantee. The result is a mechanism, not a leaderboard entry: features that name their joints are brittle to a dead joint in a way that pooled graph messages are not.

III. METHODOLOGY

We formulate exercise assessment as an *invariant read-out of an equivariant representation of an irregularly sampled pose sequence*. Two structural properties hold by construction rather than by fitting:

- (i) **Exact viewpoint invariance.** The predicted score is provably unchanged under any rigid motion of the camera. This is a theorem about the architecture (§III-C–§III-E), certified numerically to machine roundoff, and it is the paper’s central claim.
- (ii) **Grid-free consumption of arrival times.** The model reads the sensor’s actual inter-arrival intervals Δt_k and never assumes a uniform frame axis, so no resampling operator is applied to its input. We are careful to state what this does *not* buy: on KIMORE it confers no accuracy or stability advantage over a fixed-grid baseline (§IV-D). It is an architectural property that we characterise, not a performance claim that we assert.

Throughout, J is the number of skeletal joints, $g = (R, t) \in SE(3)$ acts on joint observations $x \in \mathbb{R}^{3 \times J}$ by $g \cdot x = Rx + t\mathbf{1}^\top$, and $\{t_k\}_{k=0}^{L-1}$ are the sensor’s own, non-uniform arrival stamps.

A. Irregular Arrivals Without a Frame Grid

A Kinect stream delivers observations at authentic, non-uniform times recorded by the sensor clock (the `RelativeTime` field; our anchor recording has coefficient of variation $C_V = \sigma_{\Delta t} / \mu_{\Delta t} = 0.25$ with genuine multi-frame dropouts). We first remove global body placement by referring every joint to the root,

$$\tilde{x}_{t_k} = x_{t_k} - x_{t_k}^{\text{root}} \in \mathbb{R}^{3 \times J}, \quad (1)$$

which makes translation invariance exact and discharges it *upstream of the network*. We are explicit that the encoder of §III-B is *not* itself translation-invariant – it reads joint radii $\|\tilde{x}_j\|$ and seeds vector channels from $\tilde{x}_j$ – so the claim is true only with respect to the root, which is exactly where (1) establishes it and where our certificate (E5) tests it, on raw world coordinates.

The temporal geometry enters through the inter-arrival intervals themselves,

$$\Delta t_k = t_k - t_{k-1}, \quad \Delta t_0 \equiv 0, \quad (2)$$

which are supplied to the recurrence of §III-D as an input channel. There is no frame axis to align and no uniform grid to assume: a fixed-grid architecture must first apply a resampling operator $\mathcal{R}$ (§III-G), whereas here $\mathcal{R} = \text{Id}$.

B. The $SO(3)$ -Equivariant Spatial Encoder

Each frame is encoded independently by a shared, steerable message-passing network on the skeleton graph, whose edges $\mathcal{E}$ are the anatomical bones. Per joint, the latent carries a direct sum of $O(3)$ irreducible representations — n_0 Type-0 (even scalar, $0e$) and n_1 Type-1 (odd vector, $1o$) channels,

$$h_j \in n_0 \cdot (0e) \oplus n_1 \cdot (1o), \quad h_j = (s_j, v_j), \quad (3)$$

$$s_j \in \mathbb{R}^{n_0}, \quad v_j \in \mathbb{R}^{n_1 \times 3},$$

on which the group acts by the block-diagonal Wigner- D representation

$$\rho(g) = I_{n_0} \oplus (I_{n_1} \otimes D_1(g)), \quad D_1(g) \in SO(3), \quad (4)$$

in the real spherical-harmonic basis. Every block is *orthogonal*, $D_\ell(g)^\top D_\ell(g) = I$, which we certify to $\max_g \|\rho(g)^\top \rho(g) - I\|_F \leq 10^{-12}$ in double precision.

a) *Initialisation.*: Scalars are seeded from invariants (joint radius and a learned joint embedding e_j), and vectors from the joint’s own position scaled by learned *scalars* — a scalar times a 1o feature is again 1o, hence equivariant:

$$s_j^{(0)} = \text{MLP}(\|\tilde{\mathbf{x}}_j\|, e_j), \quad v_j^{(0)} = \gamma \otimes \tilde{\mathbf{x}}_j, \quad \gamma \in \mathbb{R}^{n_1}. \quad (5)$$

b) *Steerable message passing.*: Geometry enters only through the relative bone vector $\mathbf{r}_{ij} = \tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j$ (a 1o quantity), its length $d_{ij} = \|\mathbf{r}_{ij}\|$ (an invariant), and the spherical harmonics $Y_{ij} = Y^{\ell \leq \ell_{\max}}(\mathbf{r}_{ij}/d_{ij})$. Edge weights are produced by a radial network of *invariants* only, $w_{ij} = \varphi(\text{RBF}(d_{ij}), \mathbf{b}_{ij})$ with $\mathbf{b}_{ij}$ a learned bone-type embedding, so the weights themselves are group-invariant. Messages are formed by a Clebsch–Gordan-constrained tensor product and aggregated:

$$\begin{aligned} \mathbf{m}_{ij} &= \mathcal{T}_{w_{ij}}(h_j, Y_{ij}), \\ h_i &\leftarrow h_i + W \left[\Gamma \left(\sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ij} \right) \right]. \end{aligned} \quad (6)$$

The tensor product is supported only on admissible paths $|\ell_1 - \ell_2| \leq \ell_{\text{out}} \leq \ell_1 + \ell_2$ with $p_1 p_2 = p_{\text{out}}$, and the Wigner–Eckart theorem then guarantees, for *every* admissible weight and *every* g ,

$$\mathcal{T}_w(\rho(g)h_j, D(g)Y_{ij}) = \rho(g) \mathcal{T}_w(h_j, Y_{ij}). \quad (7)$$

The gate Γ is an equivariant nonlinearity: $\tanh$ acts on feature scalars while sigmoid-activated *gate* scalars multiply the 1o channels; since a scalar commutes with $D_1(g)$, Γ preserves (7), as does the CG-constrained linear map W . Equivariance therefore holds for arbitrary weights — it is a property of the construction, not of training.

Proposition 1 (Elimination of non-scalar biases). *A learnable additive bias b on any 1o channel breaks equivariance unless $b = \mathbf{0}$; biases are admissible only on 0e channels.*

Proof. Under g a 1o feature transforms as $v \mapsto D_1(g)v$, while a constant bias transforms trivially. Equivariance of $v \mapsto v + b$ requires $D_1(g)v + b = D_1(g)(v + b)$ for all g , i.e. $D_1(g)b = b$ for all $g \in SO(3)$; the only vector fixed by the full rotation group is $b = \mathbf{0}$. For 0e channels $D_0(g) = 1$ and the constraint is vacuous. $\square$

Accordingly every steerable linear layer is instantiated with `biases=False`, and we verify structurally that no bias parameter exists on any non-scalar output.

C. The Invariant Cut: Where the Theorem Is Discharged

The encoder is *equivariant*; the score must be *invariant*. The transition is made by an explicit projection Π onto a generating set of invariants of the latent (3). The *parity-even* generators are the scalars, the norms of the vector channels, and their pairwise angles. We use unit-normalised inner products (cosines) and $\log(1 + \cdot)$ magnitudes rather than raw dot products, which are quadratic in feature scale and numerically destabilise the read-out at no informational gain.

Those generators alone, however, are complete for the *full* orthogonal group $O(3)$, not for the $SO(3)$ we require — every one of them is fixed by a reflection. A cut built from them is therefore invariant to a global mirror of the performance, and consequently blind to *trajectory handedness*: a clockwise and an anticlockwise limb circle have identical bone lengths, identical pairwise distances and identical speeds at every instant, and no parity-even feature can separate them. Any pathology whose signature is the *sense* of a rotation — compensatory trunk torsion, a circumducting gait, the direction of a scapular hitch — lies entirely in the subspace such a cut deletes. We therefore complete the cut with the *parity-odd* generators, the pseudo-scalar triple products, taken over unit vectors so that they are bounded in $[-1, 1]$ exactly as the cosines are:

$$\pi_j = \left[\underbrace{\tanh s_j \parallel \log(1 + \|v_j^c\|)_c \parallel \langle \hat{v}_j^c, \hat{v}_{j'}^{c'} \rangle_{c < c'}}_{\text{parity-even (0e)}} \parallel \underbrace{\det [\hat{v}_j^a, \hat{v}_j^b, \hat{v}_j^c]_{a < b < c}}_{\text{parity-odd (0o)}} \right], \quad (8)$$

with $\hat{v} = v/\|v\|$. These are pooled over joints by concatenated mean and max and concatenated with the invariant bone lengths $d_{ij} = \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|$ and with a fixed set $\mathcal{T}$ of *anatomical* signed volumes χ_τ — normalised determinants over physically meaningful joint triads and kinematic chains (the shoulder–shoulder–head frame, the pelvis, the signed volume of three consecutive bones along each limb). The latter do not rely on the encoder learning to construct chirality for itself:

$$\begin{aligned} \Pi(h, \tilde{\mathbf{x}}) &= \left[\text{mean}_j \pi_j \parallel \max_j \pi_j \parallel \right. \\ &\quad \left. (d_{ij})_{(i,j) \in \mathcal{E}} \parallel (\chi_\tau)_{\tau \in \mathcal{T}} \right]. \end{aligned} \quad (9)$$

a) *Structural specification of the invariant sets.*: Table I states the two sets explicitly — their generators, $O(3)$ irrep types, dimensions, and pooling — for the deployed channel widths ($n_0 = 32$ type-0, $n_1 = 8$ type-1). One point is worth making precise because it is easy to mis-specify: the parity-even and parity-odd per-joint generators are pooled by the *same* permutation-invariant mean $\oplus$ max over joints; there is no separate “signed” pooling operator for the chiral channels, and none is needed, because a mean or max of bounded signed pseudo-scalars is itself a valid invariant. The per-bone lengths and the per-frame anatomical volumes are appended *unpooled*, in fixed index order (this index-preservation is what Remark 2 relies on). Under the masked variant, a liveness gate zeroes any generator — pooled feature, bone length, or signed volume — that touches a failed joint, which is what makes the dead-node invariance of Sec. IV-E exact rather than approximate. The parity-even subtotal is exactly the 160-dimensional projection whose precision budget is audited in Sec. V-A; the parity-odd completion adds 123 dimensions and narrows the guaranteed group from $O(3)$ to $SO(3)$ at no cost to the viewpoint theorem (Remark 1).

b) *Numerical stability of the pseudo-scalars.*: The parity-odd entries divide by a vector norm, and a determinant is discontinuous in the sign of a vanishing argument, so the

TABLE I
STRUCTURAL SPECIFICATION OF THE INVARIANT CUT Π ((8)–(9)) AT THE
DEPLOYED WIDTHS $n_0 = 32$, $n_1 = 8$; $J = 25$ JOINTS, 24 BONES,
 $|\mathcal{T}| = 11$ ANATOMICAL FRAMES. BOTH PARITY CLASSES SHARE ONE
mean $\oplus$ max JOINT POOL; BONE LENGTHS AND SIGNED VOLUMES ARE
CONCATENATED INDEX-PRESERVED (UNPOOLED).

Generator	Irrep	Count	Dim
<i>Parity-even</i> — generating set complete for $O(3)$			
scalars $\tanh s$	0e	n_0	32
vector norms $\log(1 + \ v\)$	0e	n_1	8
cosines $\langle \hat{v}, \hat{v}' \rangle$	0e	$\binom{n_1}{2}$	28
<i>pooled</i> mean $\oplus$ max		2×68	136
bone lengths d_{ij}	0e	per bone	24
even subtotal			160
<i>Parity-odd</i> — $SO(3)$ chiral completion			
triple products $\det[\hat{v}, \hat{v}', \hat{v}'']$	0o	$\binom{n_1}{3}$	56
<i>pooled</i> mean $\oplus$ max		2×56	112
anatomical volumes χ_τ	0o	per frame	11
odd subtotal			123
full $SO(3)$ cut			283

unit normalisation $\hat{v} = v/\|v\|$ is regularised by a fixed floor, $\hat{v} = v/(\|v\| + \varepsilon)$ with $\varepsilon = 10^{-8}$, applied identically to the learned 1o channels and to the anatomical difference vectors. We do not clamp or skip terms; the floor is the only intervention. It is inert except when $\|v\| \lesssim \varepsilon$ — a numerically null channel or a collapsed anatomical triad — because the determinant is taken over *unit* vectors and is therefore bounded in $[-1, 1]$ regardless of ε , so the read-out cannot be inflated by a small denominator. The one regime that genuinely produces a near-zero norm is a failed tracking node that freezes two joints onto each other; there the anatomical volume is not merely small but sign-meaningless, and we zero it explicitly with a per-frame liveness mask (a chiral frame spanning a dead joint contributes exactly 0 rather than an ill-conditioned ± 1) rather than let the floor decide its sign. Because the pseudo-scalars are bounded and the floor only regularises a vanishing denominator, the deployed read-out is insensitive to ε over many decades: the invariance gate E8 below (2.9×10^{-16}) and every accuracy number in Section IV are computed with this floor active.

Proposition 2 ($SO(3)$ -invariance of the read-out). $\Pi(\rho(g)h, R\tilde{x}) = \Pi(h, \tilde{x})$ for every proper rotation $g \in SO(3)$. Consequently every downstream computation — and hence the predicted score — is invariant, for arbitrary weights. Under an improper g the parity-odd entries change sign, so the read-out is not invariant to reflections.

Proof. Scalars carry $D_0(g) = 1$ and are fixed. Vector channels transform as $v^c \mapsto D_1(g)v^c$ with $D_1(g)$ orthogonal, so $\|v^c\|$ is fixed and $\langle D_1\hat{v}^c, D_1\hat{v}^{c'} \rangle = \langle \hat{v}^c, \hat{v}^{c'} \rangle$. Bone lengths are norms of differences of positions and are fixed by any rigid motion. For the triple products, $\det[D_1\hat{v}^a, D_1\hat{v}^b, D_1\hat{v}^c] = \det(D_1) \det[\hat{v}^a, \hat{v}^b, \hat{v}^c]$, and $\det D_1(g) = +1$ precisely when g is proper; the same identity gives the sign flip for improper g . Mean and max are permutation-symmetric functions of invariants, hence invariant. $\square$

Two consequences deserve emphasis, because both are commonly mis-stated.

Remark 1 (Restoring chirality costs the viewpoint theorem nothing). *A camera relocation induces a proper rotation, for which $\det D_1(g) = +1$. The pseudo-scalars are therefore exactly invariant under every transformation the viewpoint claim quantifies over, and narrowing the guaranteed group from $O(3)$ to $SO(3)$ forfeits none of it. This is not an argument but a measurement: gate E8 of Section III-E certifies read-out invariance at 2.9×10^{-16} after the parity-odd channels are admitted, and the measured cross-viewpoint degradation is unchanged at machine precision (Section IV-C). The price is 16.7% more parameters and, on KIMORE, no measurable change in accuracy — the parity-odd subspace carries no score-relevant signal in a corpus of healthy-form exercises. We report that null rather than suppress it. The point is that the architecture no longer assumes it.*

Remark 2 (Reflection-blindness is not left/right-blindness). *It is tempting to read $O(3)$ -invariance as an inability to distinguish a movement performed with the left limb from the same movement performed with the right — which, for unilateral pathology, would be disqualifying. It does not follow, and it is false. Joint indices are preserved throughout the pipeline: the embedding of (5) is keyed by joint identity, bone lengths are emitted per bone in a fixed order, and the speed channels of (10) are indexed per joint and never pooled. A left-arm raise and a right-arm raise are not related by a reflection; they are related by a reflection composed with the left/right relabelling of joint indices, and no model here is invariant to that relabelling. Gate E7 measures it directly: swapping the left and right joint indices moves the score by 4.5×10^{-2} (parity-even cut) and 6.2×10^{-2} (full cut), while a pure reflection moves the parity-even score by exactly 0. The defect the pseudo-scalars repair is handedness of the trajectory, not identity of the limb.*

D. Temporal Recurrence on Actual Inter-Arrival Times

The encoder is a purely *spatial*, static map: it describes each frame’s pose but never its motion. We therefore hand the recurrence two families of explicitly invariant motion channels, computed from the actual stamps. For joint j at arrival k , the displacement and the true velocity magnitude are

$$\delta_{k,j} = \|\tilde{\mathbf{x}}_j(t_k) - \tilde{\mathbf{x}}_j(t_{k-1})\|, \quad \sigma_{k,j} = \frac{\delta_{k,j}}{\max(\Delta t_k, \epsilon)}. \quad (10)$$

Both are norms of differences of positions, hence $SO(3)$ -invariant by construction: they cost the theorem nothing, and the certificate (E2) re-verifies this rather than assuming it. Dividing by the *actual* Δt_k — not by a nominal frame period — is what makes σ a velocity rather than a per-frame increment, and it is the point at which irregular sampling enters the model as information instead of as a defect to be interpolated away. We include these channels because a recurrence trained on 45 subjects does not reliably learn to difference its own inputs; the ablation without them is reported in Sec. IV.

The per-arrival input is the concatenation of the invariant pose code, the interval, the motion channels, and a one-hot exercise identifier:

$$u_k = \left[\Pi_k \parallel \Delta t_k \parallel \log(1+\delta_k) \parallel \log(1+\sigma_k) \parallel \text{onehot}(e) \right], \quad (11)$$

which a bidirectional GRU consumes in arrival order. The sequence is reduced by a length-masked mean over the n genuine arrivals (padding excluded), and an MLP head emits the score:

$$\hat{y} = \psi \left(\frac{1}{n} \sum_{k < n} \text{GRU}(u_k) \right). \quad (12)$$

Since every component of u_k is invariant by Proposition 2 and (10), and ψ acts on invariants, the composite map $\tilde{x} \mapsto \hat{y}$ is exactly $SE(3)$ -invariant — for any weights, trained or not. The recurrence is discrete and consumes no solver; there is no integrator, no step-size controller, and therefore none of the adaptive-solver failure modes analysed in §III-F.

E. Numerical Certificate

An architectural symmetry argument is worth only as much as its numerical audit, because a symmetry can be silently broken by an implementation detail — and, as E6 shows, an *unintended* symmetry can be silently acquired. We therefore certify the *deployed* model (E1–E8, `src/certify_egru.py`). Gates E1–E5 establish that the invariance we claim is present; E6–E8 establish that the invariance we claim is the one we actually have.

- E1 **Representation orthogonality:** $\max_g \|\rho(g)^\top \rho(g) - I\|_F \leq 10^{-12}$ (fp64).
- E2 **Invariance of the motion channels:** δ, σ verified invariant, not assumed.
- E3 **Rotation sweep:** the score violation is $\sim 10^{-16}$ across $\theta \in [0.05, 3.0]$ rad, and — decisively — its log-log slope against θ is +0.160, i.e. *flat*. A modelling violation grows with angle; roundoff does not.
- E4 **Precision scaling:** repeating the identical computation in single and double precision gives $r = \bar{\delta}_{\text{fp32}}/\bar{\delta}_{\text{fp64}} = 2.37 \times 10^{-7}/2.96 \times 10^{-16} = 8.0 \times 10^8$, an eight-order drop tracking the ratio of machine epsilons. A genuine symmetry violation would be precision-*independent* ($r \approx 1$).
- E5 **Translation invariance of the pipeline:** tested on raw world coordinates, $|s(\mathbf{x}+\mathbf{c})-s(\mathbf{x})|/|s(\mathbf{x})| = 2.6 \times 10^{-16}$.
- E6 **Parity (the group is $SO(3)$, not $O(3)$):** under a reflection M , the parity-even cut gives $|s(M\mathbf{x}) - s(\mathbf{x})|/|s(\mathbf{x})| = 0$ *exactly* — bit-identical output on a mirrored body, an unintended $O(3)$ -invariance — whereas the completed cut of (8) gives 3.1×10^{-2} . Because the model is $SO(3)$ -invariant, all improper transformations are equivalent up to a rotation, so a single reflection certifies the entire coset.
- E7 **Laterality:** permuting the left and right joint *indices* (a relabelling, not a reflection) moves the score by 4.5×10^{-2} and 6.2×10^{-2} for the two cuts respectively. Both models separate a left-limb from a right-limb movement (Remark 2).

E8 **The fix is free:** read-out invariance under Haar-random *proper* rotations — arbitrary $g \in SO(3)$ drawn uniformly with rotation angle up to π , not a gravity-axis azimuth sweep — re-measured with the parity-odd channels active, is 2.9×10^{-16} — unchanged. Restoring chirality costs the viewpoint theorem nothing. Gates E1–E5 are likewise evaluated over Haar-random $SO(3)$, so the certified invariance is to the *full* rotation group, of which a camera azimuth or a cross-view camera displacement is one instance.

The three conditions of E3–E4 together — violation at round-off, *flat* in θ , and $r \gg 1$ — are what upgrade “small” to “zero”. Magnitude alone would not: only the flatness and the precision scaling exclude a small modelling violation masquerading as noise. E6 is included as a matter of method: an architecture should be audited for the symmetries it possesses, not only for the ones it advertises, and a gate that merely confirmed our intended invariance would have passed a model that was invariant to twice the group we wanted.

F. Control: The Continuous-Flow (Neural CDE) Variant

The natural continuous-time generalisation of §III-D replaces the discrete recurrence with a controlled differential equation driven by a spline through the arrivals. We implement it, certify it, and report it as a **control** — not as the proposed method — for a reason given empirically in Sec. IV-G: it satisfies an equally exact equivariance certificate yet *fails* the mean-predictor floor, and is therefore the cleanest available demonstration that a structural guarantee is worthless without measured signal. Its role here is to delimit the optimisation boundary of rigidly constrained continuous flows, not to carry a claim.

a) *Control path.:* The root-relative observations are interpolated by a natural cubic spline $X : [t_0, t_{L-1}] \rightarrow \mathbb{R}^{3J}$, whose second-derivative moments $M_i = X''(t_i)$ solve, with $h_i = t_{i+1} - t_i$,

$$\begin{aligned} & h_{i-1}M_{i-1} + 2(h_{i-1} + h_i)M_i + h_iM_{i+1} \\ &= 6 \left(\frac{\tilde{\mathbf{x}}_{t_{i+1}} - \tilde{\mathbf{x}}_{t_i}}{h_i} - \frac{\tilde{\mathbf{x}}_{t_i} - \tilde{\mathbf{x}}_{t_{i-1}}}{h_{i-1}} \right), \quad M_0 = M_{L-1} = \mathbf{0}. \end{aligned} \quad (13)$$

The spline is *linear* in its control points and *reproduces constants*, so it commutes with the group action, $\mathcal{S}[g \cdot \tilde{\mathbf{x}}] = g \cdot \mathcal{S}[\tilde{\mathbf{x}}]$, and is defined for arbitrary spacing h_i . The latent then evolves as

$$\begin{aligned} Z(s) &= Z(t_0) + \int_{t_0}^s f_\theta(Z(\tau), U(\tau)) d\tau, \\ U(s) &= \frac{1}{J} \sum_{j=1}^J \frac{d}{ds} X_j(s) \in \mathbb{R}^3, \end{aligned} \quad (14)$$

with f_θ a steerable field built from the same CG-constrained tensor products and gates as (6), and an invariant terminal read-out $\hat{y} = h_\psi(Z(t_{L-1}))$. For a fixed-step explicit integrator, $Z_{g \cdot X}(s) = \rho(g)Z_X(s)$ holds stage-by-stage to roundoff, independent of step size; we measure $\delta_{\text{eq}} \leq 10^{-13}$ on the real steerable architecture.

b) *The adaptive-solver failure mode, and the invariant norm.*: Fixed-step exactness is necessary but not sufficient, because a production integrator is *adaptive*. A Dormand–Prince step accepts or rejects on a scalar reduction of the embedded local error, and the accepted step sequence — hence the trajectory — is a deterministic function of that scalar. If the scalar is not group-invariant, the rotated and unrotated integrations receive *different* step grids and silently de-align. This survives an orthogonal representation, which is what makes it subtle.

Proposition 3 (Non-invariance of the per-component norm). *Let $\rho(g)$ be orthogonal and the field intertwining, so $e \mapsto \rho(g)e$. The standard component-weighted RMS norm $\mathcal{N}_W(e) = n^{-1/2} \|We\|_2$, $W = \text{diag}(w_i)$ with $w_i = (\text{atol} + \text{rtol} \cdot \max(|y_i^0|, |y_i^1|))^{-1}$, is invariant for all e if and only if $[\rho(g), W^2] = 0$. For $\text{rtol} > 0$ this fails.*

Proof. $\mathcal{N}_W(\rho(g)e)^2 = n^{-1} e^\top \rho(g)^\top W^2 \rho(g) e$, so equality for all e is $\rho(g)^\top W^2 \rho(g) = W^2$, i.e. $[\rho(g), W^2] = 0$ by orthogonality. Within a 1o block, $D_1(g)$ genuinely rotates the three components while their weights w_a, w_b, w_c are generically distinct; $[D_1(g), \text{diag}(w_a, w_b, w_c)] = 0$ for all g forces $w_a = w_b = w_c$, which per-component weighting violates whenever the components differ in scale. $\square$

Empirically this is exactly what we observe under the default norm: the accepted grids diverge ($D_{\text{grid}} \approx 40\text{--}90$ knots) and the drift degrades to $\mathcal{O}(10^{-5})$ despite an exactly equivariant field. We restore invariance by making both the tolerance and the reduction respect the irrep block structure. With F irrep features of dimension $d_f = 2\ell_f + 1$, set a block-constant tolerance from the *invariant* block norm and reduce isotropically:

$$\text{denom}_f = \text{atol} + \text{rtol} \cdot \max(\|y_f^0\|_2, \|y_f^1\|_2),$$

$$\boxed{\mathcal{N}_{\text{eq}}(\hat{e}) = \sqrt{\frac{1}{F} \sum_{f=1}^F \frac{1}{d_f} \|\hat{e}_f\|_2^2}} \quad (15)$$

with $\hat{e}_f = e_f / \text{denom}_f$.

Proposition 4 (Exact invariance and vanishing grid divergence). *$\mathcal{N}_{\text{eq}}$ is invariant; consequently the adaptive controller makes identical accept/reject decisions for X and $g \cdot X$, and $D_{\text{grid}} \equiv 0$.*

Proof. Each block transforms as $e_f \mapsto D_{\ell_f}(g)e_f$ with $D_{\ell_f}(g)$ orthogonal, so $\|y_f^k\|_2$ — and hence denom_f — is invariant, and $\|D_{\ell_f}(g)\hat{e}_f\|_2 = \|\hat{e}_f\|_2$. Thus (15) is a g -independent weighted sum of invariants. Equivalently the block-constant tolerance makes $W_f \propto I_{d_f}$, so $[D_{\ell_f}(g), W_f^2] = 0$ and the condition of Proposition 3 holds exactly. The reduced error scalar therefore coincides for X and $g \cdot X$, the step-size updates coincide, and the accepted grids satisfy $\mathcal{T}(g \cdot X) = \mathcal{T}(X)$. $\square$

Under $\mathcal{N}_{\text{eq}}$ the measured grid divergence is identically zero, the residual drift collapses to $\approx 1.2 \times 10^{-15}$ (flat in rotation magnitude), and the precision-scaling ratio is $r \approx 5 \times 10^8$. The continuous-flow control is thus *certified to the same standard as the proposed model* — which is precisely what makes

its failure against the floor (Sec. IV-G) methodologically instructive rather than merely disappointing.

G. Spectral Fidelity Calibration: The Metric $\mathcal{I}_{\text{loss}}$

A fixed-grid architecture cannot consume $(t_k, \tilde{x}_{t_k})$ directly; it must first apply a resampling operator $\mathcal{R}$ onto a uniform grid. $\mathcal{R}$ is a low-pass operation, and we quantify the high-frequency energy it removes with a *spectral fidelity calibration metric*,

$$\mathcal{I}_{\text{loss}}(\mathcal{R}) = \|\widehat{\mathcal{R}\tilde{x}}(\omega) - \widehat{\tilde{x}}(\omega)\|_{L^2(\omega > \omega_c)}^2, \quad (16)$$

evaluated above a cutoff ω_c in the tremor/jerk band. Calibrated on the real anchor with a physiologically grounded 5 Hz, 6 mm tremor injected at the distal joints, a 30% burst-drop regime already forces $\mathcal{I}_{\text{loss}} = 39\%$ under linear interpolation and 53% under cubic, rising monotonically to 81% at a 70% drop rate. Our formulation sets $\mathcal{R} = \text{Id}$ and hence $\mathcal{I}_{\text{loss}} \equiv 0$ by construction.

a) *What this metric does and does not license.*: $\mathcal{I}_{\text{loss}}$ is a statement about *signal*, not about *prediction*. It certifies that resampling provably removes a specific band of spectral energy; it does *not* certify that the removed band is *predictive* of the clinical score. We introduce it as a calibration instrument precisely so that this distinction can be tested rather than assumed — and in Sec. IV-D the test comes back negative: on KIMORE, where rehabilitation motion is slow and smooth relative to the sampling rate, a fixed-grid baseline suffers a large $\mathcal{I}_{\text{loss}}$ and yet loses essentially no accuracy, because the discarded band carries little task-relevant information. Indeed $\mathcal{R}$ acts there as an effective *denoiser*. We therefore state plainly what the metric establishes — the operational envelope within which grid-free modelling could matter — and we decline to infer a downstream penalty from a spectral one, as an earlier version of this work did.

H. Experimental Protocol

a) Corruption model anchored to real kinematics.:

The base signal is a real Kinect-v2 recording (KIMORE, healthy control NE_ID17/Es1; 783 frames at 30.3 fps), not a synthetic waveform. Its sensor-clock stamps are authentically irregular ($C_V = 0.25$, two real multi-frame dropouts), and we *calibrate* the corruption operators to reproduce them:

quantity	symbol	calibrated value
frame period	$\mu_{\Delta t}$	33.0 ms (30.3 fps)
thermal jitter	σ_j	0.53 ms
drop rate	π_B	1.3%
mean burst length	$1/p_{BG}$	5 frames
reference C_V	C_V^{ref}	0.250

Three published, physically motivated operators act on the *timestamps* (never fabricating positions): bursty drops via a two-state Gilbert–Elliott chain; jitter $\hat{t}_k = t_k + \eta_k + d_k$ with Gaussian thermal noise and an AR(1) drift ($\alpha = 0.9$), monotonicity enforced by projection; and a Gamma inter-arrival model with an Ornstein–Uhlenbeck-modulated mean. At the calibrated operating point the synthetic stream reproduces the anchor to within 10.5% in C_V (0.276 vs. 0.250).

b) *Hash-locked fairness.*: For each corruption level and seed the pipeline emits a single set of corrupted samples $\{(\tilde{t}_k, \mathbf{x}_k)\}$, and *both* architectures consume those very bytes. Ours reads the actual stamps ($\mathcal{R} = \text{Id}$); the fixed-grid baseline first applies a *best-case* operator $\mathcal{R} \in \{\text{linear}, \text{cubic}, \text{forward-fill}\}$ and we report whichever is strongest for it at every level. We compute a SHA-256 digest at each pathway’s input gate and assert byte-identity across the sweep, so no objection that the loader fed the two pathways subtly different data is available.

c) *Equal augmentation budget.*: Where a training-time robustness intervention is used (Sec. IV-D), it is drawn from a *single shared* corrupted stream that is byte-identical across arms (per-fold digests asserted equal). Augmenting only the proposed model would be the sabotaged-baseline objection with its sign reversed. Models train on drops $\sim \mathcal{U}[0, 0.3]$ and are *tested* to 0.7, so the hardest levels are genuine extrapolation rather than a memorised attack. Rotations, by contrast, are *never* applied at training time: a flat cross-viewpoint curve obtained under rotation augmentation would prove only that we had augmented.

d) *Two metrics, and why both are mandatory.*: For each method m and corruption level π we report degradation against the method’s *own* clean score, and accuracy against the clinical label,

$$\begin{aligned} \text{Degr}(m, \pi) &= |s_m(\pi) - s_m(0)|, \\ \text{MAD}(m, \pi) &= |s_m(\pi) - y|, \end{aligned} \quad (17)$$

so no cross-method scaling can bias the comparison. Degradation alone is *not* sufficient and we refuse to report it alone: **a model that ignores its input has degradation identically zero and would “win” every robustness plot ever drawn.** Accuracy-versus-level is what makes the claim mean anything, and it is interpretable only because the pooled models provably clear the mean-predictor floor. We therefore gate every experiment on that condition — the gate is an assertion in the evaluation code, not an appeal to the reader’s trust — and we draw the floor on every figure. The continuous-flow control of §III-F is the reason this gate exists: it is exactly a model with a perfect certificate, a perfectly flat curve, and no signal.

The converse failure is equally real and is why degradation is reported *alongside* accuracy rather than in place of it. A rotation-augmented baseline has a flat cross-viewpoint *accuracy* curve (Sec. IV-C) yet a per-sequence degradation of 3.03 MAD: an individual patient’s score moves by three points of fifty when the camera is moved, and the errors merely cancel in the aggregate. Aggregate accuracy cannot see this; a clinical instrument is required to give the same patient the same score from a different room. Neither metric is sufficient alone, in either direction.

e) *Training is nondeterministic; every number is a mean over three seeds.*: Our training is not bit-reproducible even at a fixed seed. The cuDNN GRU backward pass and the *e3nn* encoder’s scatter-add (`index_add_`) both accumulate with non-deterministic atomics, so identical configurations diverge run to run. We measured the resulting spread rather than assuming it away: repeated fits of the same (seed, fold) differ

by ≈ 0.33 MAD — **larger than every clean-accuracy gap between the leading models in this paper.** Reporting a single run would therefore have manufactured a result. Every accuracy figure below is consequently a mean $\pm$ standard deviation over *three* independent training seeds ($\times$ five subject-disjoint folds), and no accuracy difference smaller than this floor is interpreted as a difference. Checkpoints are keyed by (variant, seed, fold); an earlier scheme omitted the variant and an ablation silently overwrote the headline model. We note that the structural claims are *exempt* from this caution by construction: Proposition 2 and gates E1–E8 hold for arbitrary weights, so the degradation columns are weight-independent and no amount of seed noise can move them.

f) *Efficiency.*: Parameter counts are measured, not quoted: the fixed-grid transformer baseline carries 4,914,609 parameters (21.0 MB), the proposed model 565,881 (2.3 MB) — **8.7 $\times$** fewer parameters and a **9.2 $\times$** smaller memory footprint — rising to 660,345 (7.4 $\times$ fewer) once the parity-odd channels of (8) are admitted. The hand-crafted invariant control is smaller still ($\sim 0.21\text{M}$). The efficiency claim is thus retained without argument: the structural guarantees are obtained at a fraction of the baseline’s capacity, not by spending more of it.

g) *Two disclosures about the input.*: First, the loader caps every recording at 150 frames by *uniform index subsample* while preserving its true duration. A median 22.8 s KIMORE recording therefore reaches every model at ≈ 6.6 Hz (Nyquist ≈ 3.3 Hz), *not* the sensor’s 30 Hz. This is a deliberate cost control, it is applied identically to every arm, and it is load-bearing for the interpretation of Sec. IV-D: the input is already band-limited well below any frequency at which the resampling operator $\mathcal{R}$ is lossy. Second, all frequency-domain analyses in Sec. IV-D are therefore run on the *unsubsampled* 30 Hz signal, since at 6.6 Hz the band under test does not exist.

IV. RESULTS AND DISCUSSION

We report five findings, two of them negative. Section IV-A establishes what the benchmark can and cannot measure, and is a precondition for reading anything that follows. Section IV-B shows that on clean, frontal data our model, a hand-crafted invariant control, and the baseline are statistically indistinguishable — so clean accuracy is not where the case is made. Section IV-C presents the structural result: exact viewpoint invariance, held as a theorem rather than learned; the capacity tax the baseline pays to approximate it by augmentation; and the sense in which a flat accuracy curve is *not* invariance. Section IV-D reports a negative result — on KIMORE the resampling operator that fixed-grid baselines must apply is not a lossy tax but an effective denoiser — and then derives the operating condition under which that null was predictable in advance, verifying both halves of it. Section IV-E answers the question the accuracy tie leaves open: what the steerable encoder buys, if not accuracy. The answer is survival of a broken sensor, and it is measured rather than argued.

All models are trained pooled over the five exercises with subject-disjoint 5-fold cross-validation, the reported epoch se-

TABLE II
 PROTOCOL AUDIT. IDENTICAL MODEL, DATA AND SPLIT; ONLY THE EPOCH-SELECTION RULE CHANGES. *Honest* SELECTS ON A HELD-OUT VALIDATION FOLD, *inflated* TAKES THE MINIMUM OVER EPOCHS ON THE TEST SET. Δ IS THE PAIRED SUBJECT-CLUSTER BOOTSTRAP OF $\text{MAD}(\text{MODEL}) - \text{MAD}(\text{FLOOR})$.

	Exercise 1	Pooled (5 ex.)
Honest (val-selected) MAD	6.616 ± 1.476	6.065 ± 0.421
Inflated (min-on-test) MAD	5.505 ± 1.291	5.804 ± 0.390
Selection bias	+1.111 (16.8%)	+0.261 (4.3%)
Mean-predictor floor	6.699 ± 1.527	8.249 ± 1.106
Δ vs. floor	-0.081	-2.228
95% CI	$[-0.803, +0.625]$	$[-3.045, -1.429]$
Beats floor?	no	yes

lected on a held-out validation fold. Scores are in the KIMORE clinical range and MAD is reported in those units. *Floor* denotes the per-exercise mean predictor fit on the training split: a model that does not beat it has not learned subject-level skill, whatever its MAD. **Nondeterminism.** Training is not bit-reproducible: the cuDNN GRU backward pass and the *e3nn* encoder’s scatter-add both use nondeterministic atomics, and the resulting run-to-run spread (≈ 0.33 MAD) exceeds the accuracy gaps between the leading models. We therefore report every accuracy metric as mean $\pm$ std over three independent seeds, and treat any gap smaller than that spread as a tie rather than a result. The structural claim of Section IV-C is exempt from this caution by construction: it is a property of the architecture, not of any particular trained weights.

A. What the Benchmark Measures: A Protocol Audit

Published MADs on KIMORE Exercise 1 are not comparable to subject-disjoint numbers, but not for the reason one first assumes. We audited the mechanism directly and record the negative result to prevent its re-derivation: **it is not subject leakage**. In Exercise 1 each subject contributes exactly one recording, so a random `train_test_split` over samples is *already* subject-disjoint. There is no subject to leak, and the split is not at fault.

The actual mechanism is **optimistic epoch selection**. The reference protocol evaluates on the test set at every epoch and reports the minimum, with no validation split; a secondary `train+test StandardScaler` fit compounds it. Holding model, data and split fixed and changing only epoch selection isolates the bias exactly (Table II).

Two things follow, and the second matters more than the first.

(i) **The inflation is real but slice-dependent.** On Exercise 1 optimistic selection is worth +1.111 MAD (16.8%); pooled over five exercises it shrinks to +0.261 MAD (4.3%). Any comparison that mixes the two slices is meaningless, so we fix the pooled protocol for all subsequent experiments.

(ii) **On Exercise 1, the honest number is indistinguishable from the mean predictor.** The paired subject-cluster bootstrap gives $\Delta = -0.081$ with a 95% CI of $[-0.803, +0.625]$, which straddles zero. The baseline, measured honestly, does not beat a model that ignores its input entirely. Subject-level score regression on that slice therefore has *no signal*, and the

MAD improvements reported on it are not measuring assessment skill — a stronger and more troubling conclusion than the selection bias. Pooling the five exercises restores signal ($\Delta = -2.228$, CI $[-3.045, -1.429]$), and this is precisely why we pool: it is the only regime in which a degradation curve is interpretable at all.

This last point is a load-bearing methodological constraint, not a caveat. A model that ignores its input has *zero* degradation under any corruption and would “win” every robustness plot ever drawn. We therefore gate every experiment below on the model provably clearing the floor, report accuracy against the clinical label alongside degradation, and draw the floor on every figure.

B. Clean-Data Accuracy Is a Tie — and That Is the Point

On clean, frontal data the proposed model does not beat the baseline, and we do not claim it does (Table III). The steerable *e3nn* model (EGRU, 6.62 ± 0.08), a control that feeds *hand-crafted* $SO(3)$ invariants — bone lengths, joint radii, key inter-joint distances, per-joint speeds — into a plain GRU (InvariantGRU, 6.51 ± 0.07), and the point-cloud transformer baseline (PCT, 6.47 ± 0.20) are mutually indistinguishable: every pairwise gap is smaller than the 0.33 MAD seed spread, and the EGRU – InvariantGRU difference (+0.11) changes sign across seeds.

Two consequences shape how the rest of the section must be read. First, **the win, when it comes in Section IV-C, is not a win of *e3nn* machinery over a weaker architecture**. A GRU on six lines of hand-crafted invariant features is exactly as accurate and — as we show next — exactly as viewpoint-invariant. The advantage belongs to *building invariance in*, by whatever means, not to steerable tensor products specifically; we therefore present InvariantGRU as a co-result rather than bury it as an ablation. What then justifies the steerable machinery at all? Not accuracy, and we decline to argue the point rhetorically: the justification is deferred to Section IV-E, where it is *measured*, and where the hand-crafted arm — which names the joints it depends on — turns out to be destroyed by a single failed tracking node that the graph-structured encoder absorbs. Second, this tie is exactly why clean MAD cannot carry the paper — and, per Section IV-A, on the single-exercise slice it measures nothing anyway. The case rests on the structural axes.

Chirality is free. Table III also reports both arms with the parity-odd channels of (8) admitted, i.e. with the guaranteed invariance group narrowed from $O(3)$ to $SO(3)$. The change is invisible in accuracy — +0.12 MAD for EGRU, -0.20 for InvariantGRU, both inside the 0.33 seed floor and of opposite sign — which is the honest finding: on a corpus of healthy-form exercises, the parity-odd subspace carries no score-relevant signal. It costs 16.7% more parameters and nothing else, and in exchange the architecture no longer *assumes* that handedness is irrelevant. We adopt the chiral variant as the proposed model on that basis, not on an accuracy basis.

Efficiency. The tie is achieved at a fraction of the baseline’s size: EGRU carries 0.57M parameters to PCT’s 4.91M ($8.7\times$ fewer; the checkpoint shrinks from 21.0 to 2.3 MB, $9.2\times$), or

TABLE III

CLEAN-DATA ACCURACY (POOLED, SUBJECT-DISJOINT, MEAN $\pm$ STD OVER 3 SEEDS) AND MODEL SIZE. THE FOUR LEADING MODELS ARE STATISTICALLY TIED — EVERY PAIRWISE GAP IS INSIDE THE 0.33 MAD SEED FLOOR. $O(3)$ ROWS USE THE PARITY-EVEN CUT; $SO(3)$ ROWS ADMIT THE PSEUDO-SCALARS OF (8). THE MEAN-PREDICTOR FLOOR AND THE CONTINUOUS-FLOW CONTROL (CDE, SINGLE SEED) ARE SHOWN FOR REFERENCE.

Model	Group	MAD $\downarrow$	Params
PCT (baseline)	—	6.47 $\pm$ 0.20	4.91M
PCT + rotation aug.	—	7.47 $\pm$ 0.23	4.91M
InvariantGRU (hand-crafted)	$O(3)$	6.51 $\pm$ 0.07	0.20M
InvariantGRU (hand-crafted)	$SO(3)$	6.31 $\pm$ 0.16	0.21M
EGRU (ours, $e3nn$)	$O(3)$	6.62 $\pm$ 0.08	0.57M
EGRU (ours, $e3nn$)	$SO(3)$	6.73 $\pm$ 0.30	0.66M
CDE control (continuous flow)	$SO(3)$	8.43	0.57M
Mean-predictor floor	—	8.31 $\pm$ 0.05	—

0.66M with the parity-odd channels ($7.4\times$ fewer), and InvariantGRU is smaller still at 0.20M ($24.7\times$ fewer than PCT). Matching a 4.9M-parameter point-cloud transformer with a 0.2–0.7M-parameter invariant model is itself a statement about where the useful inductive bias lives.

C. Viewpoint Invariance Is a Theorem, Not a Fit

We rotate the *test* skeletons about the gravity axis through a camera azimuth of $0^\circ \rightarrow 180^\circ$. Crucially, rotations are **never** applied at training time: had we augmented, a flat curve would demonstrate only that we had augmented.

Before measuring, we certify. The architecture satisfies an $SO(3)$ -intertwining constraint whose violation we audit numerically (E1–E8, `src/certify_egru.py`): equivariance violation is $\sim 10^{-16}$ — machine roundoff — and, decisively, its log-log slope against rotation angle is $+0.160$, i.e. *flat*. A modelling violation grows with angle; roundoff does not. The fp32/fp64 violation ratio is 8.0×10^8 , confirming the residual is floating-point noise rather than a broken symmetry. Read-out invariance and translation invariance hold to 2.6×10^{-16} , and E8 re-certifies read-out invariance at 2.9×10^{-16} *after* the parity-odd channels are admitted: restoring chirality forfeits none of the guarantee.

The empirical consequence is exact, and it holds for *both* invariant models across all three seeds and both parity variants (Table IV). The EGRU and the hand-crafted InvariantGRU scores are unchanged at every azimuth: degradation is 0.00 to machine precision (worst case 9.1×10^{-6} MAD for EGRU in fp32, exactly 0 for InvariantGRU), with zero seed variance — as it must be, since invariance here is a closed-form property of the read-out, not a fitted one. The point-cloud baseline, which must learn its invariance from frontal training data, decays monotonically and **crosses the mean-predictor floor between 30° and 45°** : viewed from an oblique camera it becomes worse than predicting the exercise mean, reaching 10.4–10.8 MAD by 120° – 180° .

a) *The capacity tax of learned invariance.*: The baseline is not defenceless: rotation invariance can be *bought* with the standard remedy, training-time azimuth augmentation, and we grant it exactly that in a fourth arm (PCT+rot). Doing so

TABLE IV

CROSS-VIEWPOINT TRANSFER (MEAN OVER 3 SEEDS; TEST-TIME CAMERA AZIMUTH). BOTH INVARIANT MODELS ARE TRAINED *without* ROTATION AUGMENTATION; PCT IS SHOWN CLEAN AND WITH AZIMUTH AUGMENTATION (PCT+ROT). BASELINES USE THEIR *best* RESAMPLING OPERATOR OVER {LINEAR, CUBIC, FORWARD-FILL}. THE FINAL ROW IS THE WORST *degradation* — EACH METHOD AGAINST ITS OWN CLEAN SCORE — OVER THE WHOLE SWEEP, AND IT IS THE ROW THAT SEPARATES A GUARANTEE FROM A FIT.

Azim.	EGRU (ours, $SO(3)$)	InvariantGRU	PCT (clean)	PCT + rot
0°	6.73	6.31	6.47	7.47
15°	6.73	6.31	6.94	7.51
30°	6.73	6.31	7.84	7.62
45°	6.73	6.31	8.49	7.59
60°	6.73	6.31	<u>9.15</u>	7.62
90°	6.73	6.31	<u>10.13</u>	7.68
120°	6.73	6.31	<u>10.82</u>	7.63
150°	6.73	6.31	<u>10.74</u>	7.58
180°	6.73	6.31	<u>10.38</u>	7.57
worst degr.	9×10^{-6}	0	9.42	3.03

MAD; floor = 8.31. underline: PCT (clean) worse than the floor.

Seed std ≤ 0.30 (EGRU/Inv, and *identical* at every angle), ≤ 0.82 (PCT).

does flatten its viewpoint response — it no longer collapses with angle, holding near 7.6 MAD across the sweep. But the price is a **flat +1.0 MAD penalty** relative to un-augmented PCT (from 6.47 to 7.47 ± 0.23), which parks it *at* the mean-predictor floor at every angle. Forcing a fixed-frame model to learn a symmetry that a suitable architecture is simply *born with* costs, at this sample size, roughly the whole margin above the floor.

b) *A flat accuracy curve is not invariance.*: This is the distinction on which the section turns, and aggregate MAD cannot express it. PCT+rot is flat in *accuracy*, but its *degradation* — the movement of a given sequence’s own score — reaches **3.03 MAD**. Its flatness is statistical, not structural: individual patients’ scores swing by up to three points of fifty as the camera moves, and the errors merely cancel in the mean. Ours move by 9×10^{-6} . A screening instrument is required to return the *same patient the same score from a different room*, and that requirement is a statement about each prediction, not about a population average. Augmentation buys the average; only the architecture buys the guarantee. This is the precise sense in which the invariance is a theorem rather than a fit, and it is why we report degradation and accuracy side by side rather than either alone.

Note that both baselines are granted their best case at every angle: evaluated under all three resampling operators, we report whichever is strongest. The gap is therefore not an artifact of a sabotaged control, and the fairness matches the augmentation-budget discipline we impose in Section IV-D: rotation augmentation is offered to the baseline here exactly as drop augmentation is offered to both arms there.

c) *The guarantee is not an artifact of synthetic rotation.*: A skeptical reader may object that rotating test skeletons about the gravity axis is an *engineered* stressor. The identical encoder answers the same objection on a *real* cross-camera benchmark, and we wire it in here rather than defer it. On NTU RGB+D 60 Cross-View the train and test recordings originate from physically distinct cameras — setups 2 and 3 record the training split, and the 45° -displaced setup 1 records

a test split the model never sees at any angle during training (the test stream is never rotation-augmented). Trained **without any rotation augmentation**, our equivariant encoder transfers to the held-out camera at 82.98% top-1, and its cross-view logit invariance is exact — 5.8×10^{-9} in fp64, the same E1–E8 theorem carried onto a 60-class action head rather than a KIMORE regression head. The guarantee is not specific to the 45° camera displacement that defines this split, nor to azimuthal motion: the certificate (E1–E8) quantifies over Haar-random proper rotations spanning the *full* $SO(3)$ up to angle π , and the cross-view displacement is one instance of that group. Logit invariance is therefore exact under *arbitrary* proper rotations of the input, of which the benchmark’s inter-camera geometry is a special case. We state plainly what this is and is not. It is **not** an accuracy win: 82.98% sits some five points below a well-tuned ST-GCN on this split (published $\approx 88\%$ [8]), and under a recipe (causal sampling, root-relative coordinates, AdamW) that we did not tune to the published leaderboard — entirely consistent with the clean-data tie of Section IV-B, where accuracy was never the axis. What it *is* is the KIMORE result reproduced on real hardware: a model whose viewpoint invariance must be *learned* either sees the test camera’s pose in its training distribution or degrades, whereas ours carries the held-out 45° camera *for free and per prediction*, because the invariance is a property of the architecture and not of the training views. The cross-view generalisation others buy with augmentation, we hold by construction — and here on a benchmark with genuinely different cameras, not a synthetic sweep.

D. The Operational Envelope of Resampling: A Negative Result

Our second structural claim — that consuming irregular arrival times natively beats projecting them onto a uniform grid — **does not hold on KIMORE**. We report this in full, because the mechanism is instructive and because the claim is widely assumed.

We sweep Gilbert–Elliott burst drops with timestamp jitter from 0% to 70% at test time. Both methods receive the byte-identical corrupted timeline (SHA-256 asserted equal at every (fold, level, seed) cell); ours reads the actual stamps, the baseline must first apply the resampling operator $\mathcal{R}$, again at its best operator.

A first run was uninformative and we discard it as a training artifact. Trained on clean, near-uniform arrivals ($\Delta t \leq 0.026$ s) and then attacked with gaps up to 0.22 s — roughly $10\times$ outside its training support — our model collapsed through the floor ($6.43 \rightarrow 10.2$ MAD). The recurrence consumes Δt natively, but it had never learned what a large gap *means*. This is a deficiency of the training distribution, not of the architecture, and to test the structural question it must be removed.

We therefore grant **both** arms an identical augmentation budget, drawn from a single shared corrupted training stream that is byte-identical across arms (per-fold hash asserted equal). Augmenting only our method would be the “sabotaged baseline” objection with its sign reversed. Models are trained

TABLE V
IRREGULAR-SAMPLING STRESS TEST UNDER AN *identical, hash-locked* AUGMENTATION BUDGET FOR BOTH ARMS (TRAIN $\mathcal{U}[0, 0.3]$; TEST TO 0.7, I.E. THE LAST TWO ROWS EXTRAPOLATE). MEAN $\pm$ STD OVER 3 SEEDS; LOWER IS BETTER ON BOTH AXES.

Drop	Ours (EGRU)		Baseline (PCT, best)	
	MAD	degr.	MAD	degr.
0%	6.91 $\pm$.07	0.29	6.27$\pm$.28	0.80
10%	6.73 $\pm$.06	1.49	6.28$\pm$.27	0.90
20%	6.71 $\pm$.04	1.72	6.26$\pm$.28	1.00
30%	6.71 $\pm$.05	2.02	6.29$\pm$.27	1.16
50%	6.74 $\pm$.13	2.77	6.35$\pm$.27	1.37
70%	7.11 $\pm$.30	3.89	6.32$\pm$.27	1.36
Mean-predictor floor = 8.25; both methods clear it at all levels.				

on drops $\sim \mathcal{U}[0, 0.3]$ but *tested* to 0.7, so the two hardest levels are genuine extrapolation rather than a memorised attack.

The augmentation removes the collapse: our model now stays comfortably above the floor out to 70% drop, including the extrapolated levels, confirming the diagnosis. But it buys no advantage. The baseline is superior on *both* axes at *every* level (Table V): more accurate (6.26–6.35 vs. 6.71–7.11) and more stable (degradation 1.36 vs. 3.89 at 70%). Given the same budget, the grid-free model exploits it worse, not better.

1) *The Null Is Predicted, Not Merely Observed:* A negative result is only worth reporting if it is explained. We state the operating condition exactly, and then verify both of its halves.

A grid-free model and a resampled model see the *same surviving samples*; irregularity hands the former no extra data. They differ in exactly two respects: the grid-free model never fabricates values by interpolation, and it retains the true arrival times. It can therefore only win where those two things matter:

A grid-free model beats a resampled one if and only if the task depends on spectral content that $\mathcal{R}$ destroys.

(a) **KIMORE does not satisfy the condition.** A model-free spectral census of the corpus at its native 30.3 Hz settles this without reference to any checkpoint. Of the positional motion energy, 86.3% lies below 0.5 Hz and 97.7% below 2 Hz. The baseline’s uniform grid (100 frames over a median 22.8 s recording) has a Nyquist corner of 2.19 Hz, so $\mathcal{R}$ destroys just **2.2%** of the positional energy. It destroys 70% of the *velocity* band — but velocity is a first difference, i.e. a high-pass filter, and that band is dominated by Kinect’s broadband joint noise rather than by movement. $\mathcal{R}$ **therefore discards mostly noise**, which is precisely why the fixed-grid baseline is flat, and indeed mildly *helped*, under corruption. The operator the fixed-grid architecture is forced to apply turns out to protect it. The null of Table V is thus a prediction of the corpus, derivable before the experiment was run.

(b) **The mechanism nevertheless exists.** To confirm that the condition is the right one — and not a post-hoc rationalisation — we place signal in the band $\mathcal{R}$ destroys and re-test. We inject a physiologically calibrated 5 Hz tremor (the Parkinsonian band is 4–6 Hz) at a random amplitude into the wrist and hand joints, apply the identical Gilbert–Elliott corruption, and recover the amplitude two ways: by

resampling onto the uniform grid and matched-filtering, versus by fitting the tone directly on the actual irregular stamps (the Lomb–Scargle construction). The grid-free estimator tracks the true amplitude at $r = 1.000$ at every drop level from 0% to 70%, while the resampled estimator decays monotonically, $0.998 \rightarrow 0.785$. Irregular samples have no single Nyquist frequency and do not alias; a uniform grid at the same *average* rate folds the tone onto itself irrecoverably. The mechanism is real and behaves exactly as the theory requires.

Scope. We therefore claim no irregular-sampling advantage on KIMORE, and we do not attribute its absence to a defect of the method. We claim a *boundary*: grid-free modelling pays when the clinically relevant dynamics lie above the resampling corner — tremor, spasticity, clonus, or real sensor dropout on fast motion — and KIMORE, a corpus of slow prescribed exercises whose scores live below 2 Hz, sits strictly inside the region where it cannot. A note of caution about our own pipeline reinforces this: the loader subsamples each recording to 150 frames while preserving its duration, so every model here in fact sees ≈ 6.6 Hz, band-limiting the input still further. The regime that would test the claim is not reachable on this data, and we decline to assert the advantage without a corpus that can test it.

E. What the Steerable Encoder Buys: Sensor-Node Failure

Section IV-B left an obligation outstanding. If a plain GRU on hand-crafted invariants is as accurate as the *e3nn* encoder and exactly as viewpoint-invariant, then the steerable machinery is unjustified — and no amount of appeal to elegance repairs that. We discharge the obligation with a measurement rather than an argument.

A Kinect tracking node fails: it stops updating and reports a stale estimate. We model this literally, freezing k randomly chosen non-root joints at their first-frame position for the whole sequence — motion destroyed, geometry left anatomically plausible, which is the *conservative* version of the failure. All three arms receive the byte-identical corrupted sequence (SHA-256 asserted equal), no model is retrained, and results are pooled over three seeds and five folds (Table VI).

The hand-crafted arm does not degrade under a single dead joint; it **collapses through the mean-predictor floor** ($6.31 \rightarrow 9.78$ MAD at $k = 1$), and reaches 18.4 by $k = 8$. The EGRU loses 3.8 MAD over the same range and stays below the floor to $k = 4$. The mechanism is structural, and it is the one thing the two invariant models genuinely do not share: InvariantGRU’s features *name* the joints they depend on — specific bone lengths, specific inter-joint distances, per-joint speeds, all concatenated in index order — so killing a joint does not attenuate a feature, it corrupts a coordinate the recurrence has learned to trust. The EGRU pools *learned messages over the skeleton graph* by mean and max, so a dead node is diluted across 25 joints and its neighbours partially reconstruct it. This is invisible in every clean-accuracy column in this paper, which is exactly why it had to be measured separately.

We record two honest qualifications. First, **the point-cloud transformer is the most node-robust model of the three**

TABLE VI
SENSOR-NODE FAILURE: k TRACKING NODES FREEZE AT THEIR FIRST-FRAME POSITION. MAD, MEAN OVER 3 SEEDS $\times$ 5 FOLDS; IDENTICAL HASH-LOCKED INPUTS TO ALL ARMS; NO RETRAINING. UNDERLINE: WORSE THAN THE MEAN-PREDICTOR FLOOR (8.31).

Dead nodes k	EGRU (ours)	InvariantGRU	PCT
0	6.73	6.31	6.47
1	7.10	<u>9.78</u>	6.76
2	7.55	<u>11.64</u>	7.10
4	8.41	<u>15.96</u>	7.51
8	<u>10.50</u>	<u>18.37</u>	8.73
MAD lost	+3.76	+12.05	+2.27

(+2.3 MAD at $k = 8$): attention over a point cloud can simply down-weight an anomalous point. Robustness to node failure is not a property of equivariance, and we do not claim it is. Second, the EGRU is not immune — it is merely $3\times$ more graceful than the hand-crafted alternative. The correct reading is not “ours is most robust” but “*among models with an exact viewpoint guarantee*, only the graph-structured one survives a broken sensor.”

a) *The mechanism is not an artifact of the freeze model.*: The freeze-to-first-frame fault is one failure among many, so we re-run the experiment under four further operators — a *stuck-at-lag* node that reports the pose it held ten arrivals ago, *sporadic bursts* (Gaussian spikes on 15% of frames), single-axis *depth-channel noise*, and a *unilateral* variant that fails only left-limb joints — each at the same k levels, three model seeds, and hash-locked inputs (Table VII). The qualitative conclusion is invariant to the failure model: under every operator the hand-crafted arm degrades most and the graph-pooled encoder least among the two exact-invariance models, exactly as the naming-vs-pooling mechanism predicts. Two honest refinements emerge. First, **persistent** faults (freeze, lag) are far more destructive to the naming features (+12.05, +8.88 MAD) than **transient** ones (burst +2.83, axis +1.93), because a length-masked temporal mean dilutes an intermittent spike but not a permanently corrupted coordinate. Second, on the reviewer’s specific question of **laterality**: failing only left-side joints does not break the story or the parity-odd channels — the hand-crafted arm still collapses through the floor (+8.73) and the EGRU stays graceful (+3.43). If anything, concentrated unilateral loss is marginally gentler on the invariant models (the intact contralateral side aids the graph’s reconstruction) and marginally *harder* for the point-cloud transformer (+3.15 vs. +2.27 random), whose attention down-weights scattered anomalies more easily than a coherent one-sided dropout.

b) *Steerable vs. lighter $E(n)$ equivariance.*: This section admits a second control the reviewer rightly asked for: is it the *steerable* tensor-field machinery that buys node-failure survival, or would any equivariant encoder do? We swap the *e3nn* encoder for a from-scratch Satorras $E(n)$ -equivariant graph network [13] behind the *identical* invariant cut and pooling, so viewpoint invariance is a tie *by construction* — the cut makes EGNN + cut exactly $SE(3)$ -invariant, certified to 1.4×10^{-5} in fp32 — and the only free variable is the

TABLE VII

NODE FAILURE UNDER FIVE OPERATORS (CHIRAL MODELS, POOLED, MEAN OVER 3 SEEDS $\times$ 5 FOLDS; HASH-LOCKED INPUTS, NO RETRAINING). EACH CELL IS MAD AT $k=1$ / $k=8$ DEAD JOINTS AND THE TOTAL MAD LOST OVER THAT RANGE. THE FREEZE/RANDOM ROW REPRODUCES TABLE VI; EVERY OPERATOR PRESERVES THE ORDERING.

Failure operator	EGRU (ours) $k1/k8$ (Δ)	InvariantGRU $k1/k8$ (Δ)	PCT $k1/k8$ (Δ)
freeze (random)	7.10/10.50 (+3.76)	9.78/18.37 (+12.05)	6.76/8.73 (+2.27)
freeze (left-only)	7.11/10.16 (+3.43)	10.81/15.04 (+8.73)	6.83/9.62 (+3.15)
stuck-at-lag	6.82/7.76 (+1.03)	9.02/15.19 (+8.88)	6.60/7.11 (+0.64)
sporadic burst	6.74/6.78 (+0.05)	6.69/9.14 (+2.83)	6.47/6.45 (−0.02)
axis/depth noise	6.77/7.20 (+0.47)	6.70/8.24 (+1.93)	6.48/6.50 (+0.03)

Floor = 8.25; $k=0$ MAD is 6.80/6.41/6.21 for EGRU/Inv/PCT.

TABLE VIII

ENCODER EQUIVARIANCE CLASS BEHIND THE *identical* INVARIANT CUT AND POOLING (ISOLATING THE STEERABLE MACHINERY): A LIGHTER $E(n)$ (EGNN) ENCODER AGAINST THE STEERABLE $e3nn$ ENCODER AND THE HAND-CRAFTED INVARIANTS. MEAN OVER 3 SEEDS $\times$ 5 FOLDS; HASH-LOCKED INPUTS; NO RETRAINING. ALL THREE ARE EXACTLY VIEWPOINT-INVARIANT — A THEOREM FOR EGRU/INVARIANTGRU, AND AN FP32 CERTIFICATE OF 1.4×10^{-5} FOR EGNN + CUT. “NODE-FAIL” IS MAD LOST OVER 8 DEAD JOINTS. DAMPING THE EGNN COORDINATE UPDATES (CLAMP 0.1/0.5/1.0) GIVES +6.42/ +6.06/ +5.69 — THE GAP DOES NOT CLOSE.

Encoder (identical cut)	Clean MAD	Node-fail lost
Steerable $e3nn$ (EGRU, ours)	6.73	+3.76
Lighter $E(n)$ (EGNN)	6.88 ± 0.16	$+6.39 \pm 1.27$
Hand-crafted (InvariantGRU)	6.31	+12.05

encoder’s equivariance class (Table VIII). On clean data the two tie (EGNN 6.88 ± 0.16 vs. EGRU 6.73, inside the 0.33 seed floor), so the lighter encoder is not crippled. Under node failure it is: EGNN loses $+6.39 \pm 1.27$ MAD over 8 dead joints against the steerable encoder’s +3.76 — a gap of ≈ 2.6 MAD ≈ 3.6 SEM across three seeds — landing *between* the steerable encoder and the hand-crafted InvariantGRU (+12.05). Damping the EGNN coordinate updates does not close it (clamp 0.1/0.5/1.0 $\rightarrow$ +6.42/ +6.06/ +5.69; the best arm is still ≈ 2 MAD worse than EGRU), so the brittleness is intrinsic to the lighter equivariance, not an artifact of coordinate-update magnitude. We state the caveat plainly: this EGNN is a from-scratch encoder at fixed width (4 layers, hidden 64) without a hyperparameter search, so we do *not* claim “steerable spherical harmonics beat $E(n)$ ” in general. What the clean-accuracy tie and the seed and clamp controls *do* license is the scoped claim that, behind this identical cut and on this corpus, the steerable encoder is measurably the more node-fail-robust of the two — the outcome that justifies its cost.

c) *The whole thesis in one grid.*: Table IX places every arm against both stressors at once, and it is deliberately unflattering. **We do not claim to be the only survivor**: rotation-augmented PCT also clears the floor under both a 90° camera move and two dead nodes. Augmentation is a real answer to viewpoint and we will not pretend otherwise. What separates the two survivors is the final column. Both are flat in aggregate *accuracy*, but the augmented baseline’s worst per-sequence *degradation* is 3.03 MAD against our 9×10^{-6} : an individual patient’s score moves by three points of fifty

TABLE IX

DOES EACH MODEL STAY BELOW THE MEAN-PREDICTOR FLOOR (8.31 MAD) UNDER EACH STRESS? THE LAST COLUMN IS THE WORST PER-SEQUENCE SCORE SHIFT UNDER *any* CAMERA ROTATION — THE DIFFERENCE BETWEEN A GUARANTEE AND A FIT.

Model	Params	Clean	90° view	2 dead	Worst degr.
EGRU (ours)	0.66M	6.73	6.73 ✓	7.55 ✓	9×10^{-6}
InvariantGRU	0.21M	6.31	6.31 ✓	11.64 ×	0
PCT (baseline)	4.91M	6.47	10.13 ×	7.10 ✓	9.42
PCT + rot	4.91M	7.47	7.68 ✓	7.76 ✓	3.03

purely by relocating the camera, and the errors merely cancel in the population mean. A screening instrument must return the same patient the same score from a different room — a per-prediction requirement that an aggregate cannot express. Secondly, and outside the seed floor, the surviving structural model is also more accurate on clean data (6.73 vs. 7.47) and $7.4\times$ smaller.

F. Per-Family Contribution of the Invariant Cut

To quantify what each family of the cut (8) contributes, we zero one family at a time at *inference* on the deployed chiral checkpoints, preserving the projection dimension so the ablation runs without retraining (Table X). This isolates the trained model’s *dependence* on each family; it is not the marginal value of *adding* a family at training, which for the parity-odd subspace is already the $O(3)$ -vs- $SO(3)$ retrain comparison of Table III, and the two readings are reconciled below.

Accuracy. Three families carry it. Removing the pairwise *cosines* costs 1.83 MAD and the learned *triple products* 2.75, with bone lengths a distant third (0.67); the scalar, vector-norm and anatomical-volume families are each inside the 0.33 seed floor (+0.25, +0.16, +0.19) and are inert. The pairwise angular structure of the encoder’s vector channels is where the score lives.

The parity-odd families split, and that reconciles the chirality null. The two parity-odd families behave oppositely: the *learned* triple products are heavily relied upon (+2.75), while the *fixed anatomical* volumes are inert (+0.19). This is consistent with — not contrary to — the finding of Section IV-B that *adding* chirality is free (+0.11 MAD, Table III). That null is a *retrain* comparison: a model trained without the parity-odd subspace matches it, because the parity-even families substitute for it during optimisation. The present ablation is an *inference* comparison on a model trained *with* it: given learnable triple-product channels the optimiser routes signal through them, so zeroing them at test time starves weights that expect them. The fixed anatomical volumes, carrying only a handedness that healthy-form exercises never reward, are never co-opted — the chirality null, now localised to the family that embodies it. Both statements hold: the parity-odd subspace is free to add and, once added, used.

Robustness. The node-failure column (worst MAD lost over 8 dead joints, against the full cut’s +3.76) resolves the mechanism of Section IV-E to family granularity. Removing the *cosines* is doubly harmful — it inflates node-failure degradation the most (+7.29), so the angular structure carries

TABLE X

PER-FAMILY ABLATION OF THE INVARIANT CUT (8): EACH FAMILY ZEROED AT INFERENCE ON THE DEPLOYED CHIRAL EGRU (PROJECTION DIMENSION PRESERVED, *no retraining*; MEAN OVER 3 SEEDS $\times$ 5 FOLDS). ΔACC IS THE CLEAN-MAD COST OF REMOVING THE FAMILY; “NODE-FAIL LOST” IS THE WORST MAD LOST OVER 8 DEAD JOINTS (FULL-CUT BASELINE $+3.76$). ΔACC BELOW THE 0.33 SEED FLOOR IS A TIE.

Family removed	Parity	Clean MAD	ΔAcc	Node-fail lost
— (full cut)	—	6.73	—	+3.76
scalars	even	6.98	+0.25	+3.46
vector norms	even	6.89	+0.16	+3.51
cosines	even	8.56	+1.83	+7.29
triple products	odd	9.48	+2.75	−0.16
anat. volumes	odd	6.93	+0.19	+3.43
bone lengths	even	7.40	+0.67	+2.36

graceful degradation as well as accuracy. Conversely, the families that most *name* their joints are the brittleness locus: zeroing bone lengths (+2.36) or the learned triple products (−0.16) *flattens* the node-failure response, but at an accuracy cost that puts the model through the floor, which is why the cut keeps them. This is the within-model echo of the between-model result of Section IV-E: features that name joints trade robustness for the accuracy they carry.

G. Limitations

We flag five, all measured rather than anticipated.

The continuous-time variant does not work. The Neural CDE, evaluated identically, reaches 8.427 MAD against a floor of 8.249 — it *fails* the mean-predictor floor and beats it on only 3/5 folds. Its equivariance certificate is nonetheless exact, which is the cautionary point, and the reason our evaluation gates on the floor: a valid structural guarantee on a model without signal produces a perfectly flat, entirely meaningless robustness curve. We report the CDE as a control and rest no claim on it. This is also why nondeterminism, which shifts absolute MAD by up to 0.33, never threatens the central result: the flat curve is a theorem about the architecture, and it stayed exactly flat across every seed even as the accuracy anchor moved.

Clean-data accuracy is a tie, not a win. On undistorted frontal data EGRU, InvariantGRU, and PCT are statistically indistinguishable (Table III); we make no clean-accuracy claim and the case rests entirely on the structural axes. In particular we do *not* claim the *e3nn* machinery is necessary for *accuracy*: a GRU on hand-crafted invariants matches it, and on this corpus that is the more parameter-efficient way to obtain an exact viewpoint guarantee. Its justification is confined to what Section IV-E measures.

Augmentation is a viable alternative to architecture, and we say so. Rotation-augmented PCT clears the floor under both stressors (Table IX). Our claim against it is not that it fails but that its invariance is statistical rather than exact (3.03 vs. 9×10^{-6} MAD of per-sequence score shift) and that it costs +1.0 MAD and $7.4\times$ the parameters. A reader who regards a three-point-in-fifty viewpoint-dependent score swing as clinically acceptable should prefer the augmented baseline, and we would not dispute it.

The irregular-sampling advantage does not hold here, and could not have. Section IV-D1 shows the resampling operator destroys only 2.2% of KIMORE’s positional energy and acts as a denoiser; the mechanism is real (grid-free tremor recovery at $r = 1.000$ under 70% drops) but the corpus does not exercise it. The claim is deferred to a corpus with genuine high-frequency clinical content, not asserted on this one. We note that our own loader, which subsamples to 150 frames, band-limits the input further — so this paper cannot test the claim even in principle, and a follow-up must run at the sensor’s native rate.

Chirality is restored but unrewarded. The parity-odd channels close a genuine hole — the cut was silently $O(3)$ -invariant and provably blind to trajectory handedness (E6) — at no cost to the viewpoint theorem (E8). But they buy no accuracy on KIMORE (± 0.2 MAD, inside the seed floor and of inconsistent sign), because a corpus of healthy-form prescribed exercises contains no chiral pathology to detect. The fix is principled, not empirically vindicated, and we present it as the former. A corpus with unilateral or rotational pathology is required to reward it.

Augmented-training floor margin is thin on one fold. Under drop augmentation the EGRU clears the mean-predictor floor in aggregate on all three seeds (6.80–7.00 vs. 8.25–8.38), but on 1 of 15 (seed, fold) cells it does not (6.96 vs. 6.82). Degradation on that cell is reported but not interpreted, per the same gate.

V. EDGE DEPLOYMENT: PRECISION BUDGET AND STREAMING LATENCY

The viewpoint guarantee is a statement about real arithmetic; a device in a patient’s home runs reduced precision and must produce a score before the exercise is over. We close the paper by measuring what the theorem costs on edge hardware, on the *deployed* clinical model and on a causal, real-time variant of the encoder. Both are deployment questions the clean-accuracy column cannot ask.

A. Equivariance Has a Precision Budget

Quantisation is not an orthogonal transformation, so it does not commute with the Wigner- D action: the exact identity $s(g \cdot x) = s(x)$ acquires a numerical floor once the representation is stored in fewer bits. We measure it on the trained pooled EGRU over 40 real KIMORE sequences $\times$ 64 camera azimuths, constructing each arithmetic *natively* (the Clebsch–Gordan tables are built in the target format, not cast after the fact, which would report a spurious floor). We report two quantities: the relative violation of the 160-dimensional invariant projection — the true arithmetic floor, which does not saturate — and the resulting movement of the reported 0–50 score, in the MAD units a clinician reads (Table XI). The fp32 row reproduces the 9×10^{-6} MAD residual of Table IV: that number was the fp32 point of a ladder.

The floor climbs monotonically with the mantissa, spanning fourteen decades from fp64 to int8 — equivariance has a measurable precision budget. Two details matter for deployment. First, fp16 is $3.2\times$ tighter than bf16 despite equal width: bf16

TABLE XI

EQUIVARIANCE PRECISION BUDGET OF THE DEPLOYED POOLED EGRU (REAL KIMORE MOTION, 64 AZIMUTHS, NATIVE PER-FORMAT CONSTRUCTION). “INV FLOOR” IS $\max_g \|\text{inv}(g \cdot x) - \text{inv}(x)\| / \|\text{inv}(x)\|$ ON THE INVARIANT PROJECTION; “SCORE SHIFT” IS THE MOVEMENT OF THE REPORTED 0–50 SCORE. THE INT8 ROWS DECOMPOSE THE CLIFF INTO ITS WEIGHT- AND ACTIVATION-QUANTISATION PARTS.

Arithmetic	INV floor (rel.)	Score shift (MAD)	Note
fp64	5.75×10^{-16}	3.3×10^{-14}	exact (theorem)
fp32	2.68×10^{-7}	8.9×10^{-6}	= Table IV
fp16	2.62×10^{-3}	0.024	> bf16
bf16	8.34×10^{-3}	0.195	
int8, weights only	3.09×10^{-7}	1.2×10^{-5}	theorem holds
int8, activations only	3.38×10^{-2}	0.056	cliff
int8, weights+acts	3.40×10^{-2}	0.051	deployable

trades mantissa for exponent range, and equivariance pays in mantissa bits, so fp16 is the better half-precision target for this model. Second, int8 (simulated per-tensor symmetric fake-quantisation with an MSE-calibrated scale, since e3nn has no native int8 kernel) falls *off* the float curve, and decomposing it is the point: quantising the *weights* leaves the floor at its fp32 value (3.1×10^{-7}), because the encoder is equivariant for arbitrary weights — a quantised model is simply a different equivariant model. The cliff to 3.4×10^{-2} appears only when the *activations* are quantised, because rounding the type-1 (vector) channels and the rotated joints onto a fixed grid is basis-dependent: $\text{round}(\rho(g)v) \neq \rho(g)\text{round}(v)$. The structural break is an activation-quantisation phenomenon, not a weight-precision one.

Yet the *clinical* consequence stays small. Even at int8, the reported score moves by 0.05 MAD — two orders below the mean-predictor floor and $60\times$ smaller than the 3.03 MAD per-sequence swing that camera rotation inflicts on the augmented baseline (Table IV). Int8 shatters *exact* structural equivariance while leaving the deployed instrument’s output clinically stationary: the honest arithmetic number and the honest deployment number are different numbers, and we report both.

B. Causal Streaming and Time-to-First-Score

A high-accuracy skeleton classifier is often bidirectional and pools over the whole clip, so it cannot emit a score until the last frame arrives — it has no honest latency. Because our viewpoint guarantee lives entirely in a per-frame spatial encoder followed by an invariant cut, it survives conversion to a strictly causal read-out unchanged: replacing the bidirectional recurrence with a unidirectional one and the full-sequence pool with a causal running mean touches nothing left of the cut. We verify this on the NTU RGB+D 60 Cross-View action benchmark, where high-frequency motion actually exercises a streaming model, using the identical equivariant encoder trained for that task; the causal head is retrained on the frozen encoder’s features (Table XII).

The conversion costs 3.17 points of top-1 accuracy ($82.98 \rightarrow 79.81\%$) — the price of never seeing the future — and buys a genuine real-time budget: a full incremental step (encoder, invariant projection, one recurrent update, running-mean pool,

TABLE XII

BIDIRECTIONAL VS. CAUSAL-STREAMING EGRU ON NTU RGB+D 60 CROSS-VIEW. THE STREAMING VARIANT REUSES THE *same* EQUIVARIANT ENCODER; ONLY THE TEMPORAL READ-OUT CHANGES (UNIDIRECTIONAL RECURRENCE, CAUSAL RUNNING-MEAN POOL). LATENCY IS A GENUINE PER-FRAME INCREMENTAL STEP WITH THE HIDDEN STATE CARRIED, NOT A WHOLE-CLIP FORWARD PASS.

	Bidirectional	Causal streaming
Top-1 accuracy (%)	82.98	79.81
Per-frame latency	—	8.9 ms (112 fps)
Time-to-first-score	full clip	8.9 ms
Frames to lock on (median)	—	54
Viewpoint residual (fp64)	5.8×10^{-9}	1.4×10^{-8}

head) runs in 8.9 ms on a single GPU, or 112 frames per second, comfortably inside the 33 ms budget of a 30 fps sensor. The first score therefore exists 8.9 ms after the first frame. It is not yet *trustworthy* that early: read out at each frame, the causal running-mean prediction rises from chance (1.9% at one frame) through 45% at 50 frames to 80.6% at 100, and locks onto the eventual class after a median of 54 frames. We report the whole trajectory rather than a single latency, because “time to a *correct* score” and “time to a *score*” are different guarantees. Throughout, the viewpoint theorem is undamaged: the streaming logits are viewpoint-invariant to 1.4×10^{-8} in fp64, statistically the same floor as the bidirectional model’s 5.8×10^{-9} (the residual is the invariant-projection floor of Sec. V-A amplified through the recurrent head, not a symmetry violation). Real-time causal operation and the per-prediction guarantee are not in tension.

VI. CONCLUSION

We set out to beat a point-cloud transformer on rehabilitation assessment and, on the metric the field reports, we did not: on clean frontal KIMORE our model, the baseline, and a GRU on hand-crafted invariants are statistically tied, and we quantified the nondeterminism floor (0.33 MAD) that makes them tied rather than merely close. We regard establishing that as a contribution rather than a concession, because it forces the real question. Clean accuracy on this benchmark cannot separate architectures; on the single-exercise slice most commonly reported, it cannot even separate a trained model from a mean predictor once epoch selection is made honest.

What separates them is what happens off the clean, frontal manifold. Our read-out is $SE(3)$ -invariant by construction and certified to machine roundoff — flat in rotation angle, with an eight-order fp32/fp64 precision-scaling signature — so moving the camera moves a patient’s score by 9×10^{-6} MAD, for arbitrary weights. The baseline, which must learn its invariance, degrades past the mean predictor by 45° . Augmentation genuinely rescues it in the aggregate, and we say so; what it does not rescue is the individual prediction, which still swings by 3.03 MAD — three points of fifty — purely by relocating the camera. A screening instrument must return the same patient the same score from a different room. That requirement is per-prediction, and only the architecture supplies it.

We reported two nulls without softening them. Native consumption of irregular arrival times bought nothing here, and we derived the operating condition that made the null

predictable in advance: the resampling operator a fixed-grid model is forced to apply destroys 2.2% of KIMORE’s positional energy and functions as a denoiser, because a corpus of slow prescribed exercises has essentially no energy above the resampling corner. The mechanism is nonetheless real, and we verified it by putting signal where the operator is lossy: a grid-free estimator tracks a 5 Hz tremor at $r = 1.000$ through 70% frame drops while a resampled one decays to 0.785. Grid-free modelling is a claim about tremor, spasticity, clonus and fast motion under real dropout — not about this corpus, and we decline to assert it on a corpus that cannot test it. Similarly, closing the parity hole is principled but empirically unrewarded: healthy-form prescribed exercises contain no chiral pathology, so the pseudo-scalars buy nothing here beyond the removal of an assumption the architecture had been making silently.

The steerable encoder earns its place on one axis, and it is measured rather than argued. When a tracking node freezes, the hand-crafted invariant model — equally accurate, equally viewpoint-exact, three times smaller — collapses through the mean-predictor floor at a *single* dead joint, because its features name the joints they depend on. The graph encoder dilutes the same failure across the skeleton. We are careful about what this does and does not say: the point-cloud transformer is in fact the most node-robust model of the three, and robustness to node failure is not a property of equivariance. The defensible claim is the Pareto one. Among models that carry an exact, per-prediction viewpoint guarantee, only the graph-structured equivariant one survives a broken sensor — and it does so at $7.4\times$ fewer parameters than the transformer it is compared against.

Two recommendations follow for the field. First, report a mean-predictor floor and a per-sequence degradation alongside aggregate error; without the former a robustness curve is uninterpretable, and without the latter a model whose errors merely cancel is indistinguishable from one that is invariant. Second, evaluate off the frontal manifold as a matter of course — camera relocation and sensor-node failure are not adversarial constructions but the ordinary conditions of a device deployed in a patient’s home. The corpus that would reward the two capabilities we built and could not vindicate here — high-frequency clinical content at the sensor’s native rate, and unilateral or rotational pathology — is the one we intend to build next.

Code and Data Availability

KIMORE [1] is publicly available from its authors. The equivariance certificate (gates E1–E8), the corruption pipeline, the spectral census, the node-failure harness, and the aggregation script that regenerates every number in this paper from banked artifacts will be publicly released upon acceptance. The repository URL is withheld here to preserve double-blind review, and an anonymized copy is available to reviewers on request.

REFERENCES

- [1] M. Capecci, M. G. Ceravolo, F. Ferracuti, S. Iarlori, A. Monteriu, L. Romeo, and F. Verdini, “The KIMORE dataset: Kinematic assessment

- of movement and clinical scores for remote monitoring of physical rehabilitation,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 7, pp. 1436–1448, 2019.
- [2] “A point cloud transformer for remote monitoring and automated assessment of physical rehabilitation exercises,” arXiv:2606.30309, 2026.
- [3] M. Karlov, A. Abedi, and S. S. Khan, “Rehabilitation exercise quality assessment through supervised contrastive learning with hard and soft negatives,” *Med. Biol. Eng. Comput.*, 2024.
- [4] A. Abedi, M. Malmirian, and S. S. Khan, “Cross-modal video to body-joints augmentation for rehabilitation exercise quality assessment,” arXiv:2306.09546, 2023.
- [5] Z. Kuang, Y. Yin, Y. Yang, J. Zhao, and L. Sun, “Dual-stream STGCN with motion-aware grouping for rehabilitation action quality assessment,” *Sensors*, vol. 26, no. 1, p. 287, 2026.
- [6] A. Ismail-Fawaz et al., “Rehab-Pile: A cross-subject rehabilitation benchmark aggregating KIMORE, UI-PRMD and IRDS,” arXiv:2507.21018, IEEE FG 2026.
- [7] A. Vakanski, H.-P. Jun, D. Paul, and R. Baker, “A data set of human body movements for physical rehabilitation exercises (UI-PRMD),” *Data*, vol. 3, no. 1, p. 2, 2018.
- [8] S. Yan, Y. Xiong, and D. Lin, “Spatial temporal graph convolutional networks for skeleton-based action recognition,” in *Proc. AAAI*, 2018.
- [9] T. S. Cohen and M. Welling, “Group equivariant convolutional networks,” in *Proc. ICML*, 2016.
- [10] N. Thomas, T. Smidt, S. Kearnes, L. Yang, L. Li, K. Kohlhoff, and P. Riley, “Tensor field networks: Rotation- and translation-equivariant neural networks for 3D point clouds,” arXiv:1802.08219, 2018.
- [11] F. B. Fuchs, D. E. Worrall, V. Fischer, and M. Welling, “SE(3)-Transformers: 3D roto-translation equivariant attention networks,” in *Proc. NeurIPS*, 2020.
- [12] M. Geiger and T. Smidt, “e3nn: Euclidean neural networks,” arXiv:2207.09453, 2022.
- [13] V. G. Satorras, E. Hoogeboom, and M. Welling, “E(n) equivariant graph neural networks,” in *Proc. ICML*, 2021.
- [14] C. Deng, O. Litany, Y. Duan, A. Poulenard, A. Tagliasacchi, and L. Guibas, “Vector neurons: A general framework for SO(3)-equivariant networks,” in *Proc. ICCV*, 2021.
- [15] O. Puny, M. Atzmon, H. Ben-Hamu, I. Misra, A. Grover, E. J. Smith, and Y. Lipman, “Frame averaging for invariant and equivariant network design,” in *Proc. ICLR*, 2022.
- [16] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud, “Neural ordinary differential equations,” in *Proc. NeurIPS*, 2018.
- [17] Y. Rubanova, R. T. Q. Chen, and D. Duvenaud, “Latent ODEs for irregularly-sampled time series,” in *Proc. NeurIPS*, 2019.
- [18] P. Kidger, J. Morrill, J. Foster, and T. Lyons, “Neural controlled differential equations for irregular time series,” in *Proc. NeurIPS*, 2020.
- [19] Z. Che, S. Purushotham, K. Cho, D. Sontag, and Y. Liu, “Recurrent neural networks for multivariate time series with missing values,” *Sci. Rep.*, vol. 8, no. 1, p. 6085, 2018.
- [20] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, “Learning phrase representations using RNN encoder-decoder for statistical machine translation,” in *Proc. EMNLP*, 2014.
- [21] J. R. Dormand and P. J. Prince, “A family of embedded Runge–Kutta formulae,” *J. Comput. Appl. Math.*, vol. 6, no. 1, pp. 19–26, 1980.
- [22] J. T. VanderPlas, “Understanding the Lomb–Scargle periodogram,” *Astrophys. J. Suppl. Ser.*, vol. 236, no. 1, p. 16, 2018.